Building and evaluation of a PBPK model for Metoprolol in CYP2D6 phenotype and activity-score groups

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based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Metoprolol-Model/releases/tag/v1.0
OSP Version	12.2
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This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

Table of Contents

- 1 Introduction
- 2 Methods
  - 2.1 Modeling strategy
  - 2.2 Data used
  - 2.3 Model parameters and assumptions
- 3 Results and Discussion
  - 3.1 Metoprolol final input parameters
  - 3.2 Diagnostic plots
    - 3.2.1 Metoprolol racemate goodness-of-fit diagnostics
    - 3.2.2 (R)-Metoprolol goodness-of-fit diagnostics
    - 3.2.3 (S)-Metoprolol goodness-of-fit diagnostics
    - 3.2.4 α-Hydroxymetoprolol goodness-of-fit diagnostics
  - 3.3 Concentration-time profiles
    - 3.3.1 Model Building
    - 3.3.2 Model Verification
- 4 Conclusion
- 5 References

1 Introduction

Metoprolol is a selective beta1-adrenoceptor blocker used in cardiovascular indications including hypertension, angina pectoris, arrhythmias and heart failure. Metoprolol pharmacokinetics show substantial interindividual variability because CYP2D6 is the dominant enzyme involved in oxidative metabolism and CYP2D6 activity varies across genetic activity-score groups (Rüdesheim 2020).

The model describes metoprolol as a whole-body parent-metabolite PBPK model with separate (R)-metoprolol and (S)-metoprolol enantiomers, racemic metoprolol as an observed composite endpoint and α-hydroxymetoprolol as a metabolite. This structure supports evaluation of racemic and enantiomer-specific plasma concentration-time data after intravenous infusion, oral immediate-release metoprolol tartrate and oral controlled-release metoprolol succinate administration.

The clinical evaluation covers adult plasma pharmacokinetic data across a dose range of 5 to 200 mg and includes studies with CYP2D6 activity-score or phenotype information. The model was developed to describe metoprolol disposition, α-hydroxymetoprolol formation and CYP2D6 drug-gene effects, and was subsequently used as a CYP2D6 substrate model in the broader CYP2D6 drug-drug-gene interaction network (Rüdesheim 2020, Rüdesheim 2025).

2 Methods

2.1 Modeling strategy

The PBPK model was developed using the Open Systems Pharmacology Suite. Distribution and elimination processes were implemented in a whole-body model structure using standard physiological organ compartments and compound-specific physicochemical and biochemical input parameters.

Model development followed an iterative workflow in which in vitro and physicochemical information was combined with clinical plasma concentration-time data. Parameters with direct literature support were used as fixed inputs where possible, while selected parameters controlling oral absorption, CYP2D6 activity-score-dependent metabolism and residual clearance were optimized against clinical data.

The model represents (R)-metoprolol and (S)-metoprolol as separate compounds because CYP2D6-mediated metabolism is enantiomer-specific. Racemic metoprolol is represented as an observer output for comparison with clinical studies reporting total metoprolol. α-Hydroxymetoprolol is included as a measured metabolite formed through the α-hydroxylation pathway.

The base metoprolol model was informed by intravenous and oral metoprolol studies covering immediate-release tartrate and controlled-release succinate formulations. Intravenous infusion studies supported systemic disposition and clearance assumptions, while oral studies informed formulation and first-pass behavior. The CYP2D6 activity-score model used clinical studies with reported CYP2D6 phenotypes, genotypes or derived activity scores to identify activity-score-dependent k_{cat} values for both enantiomers (Rüdesheim 2020).

The major proteins represented in the model are CYP2D6 and CYP3A4. CYP2D6 accounts for α-hydroxylation and O-demethylation pathways and is activity-score-dependent. CYP3A4 is represented as a first-order clearance component to account for CYP2D6-independent oxidative metabolism. Renal filtration is included for both enantiomers and for α-hydroxymetoprolol. Unspecific hepatic clearance is included only for α-hydroxymetoprolol.

2.2 Data used

In vitro and physicochemical data

Drug-dependent input parameters for the metoprolol enantiomers were taken from the published metoprolol PBPK model and its cited primary sources (Rüdesheim 2020). The model uses the same physicochemical properties for both enantiomers when the source data do not support an enantiomer-specific difference.

Parameter	Unit	(R)-metoprolol	(S)-metoprolol	Source	Description
MW	g/mol	267.36	267.36	Literature	Molecular weight.
pK_a,base	-	9.70	9.70	Literature	pKa of a basic ionization site.
Solubility, tartrate	g/mL	1.00	1.00	Literature	Aqueous solubility of the tartrate salt at pH 7.4.
Solubility, succinate	g/mL	0.16	0.16	Literature	Aqueous solubility of the succinate salt at pH 5.5.
logP	-	1.77	1.77	Literature	Octanol-water partition coefficient of the neutral species.
f_u	%	88	88	Literature	Fraction unbound in plasma.
$K_{m,CYP2D6}$, αHM	µmol/L	10.08	10.75	Mautz 1995	Michaelis constant for α-hydroxymetoprolol formation.
$k_{cat,CYP2D6}$, αHM, AS=2	1/min	10.17	11.19	Optimized	Catalytic rate constant for α-hydroxymetoprolol formation.
$K_{m,CYP2D6}$, ODM	µmol/L	8.82	12.43	Mautz 1995	Michaelis constant for O-demethylmetoprolol formation.
$k_{cat,CYP2D6}$, ODM, AS=2	1/min	16.69	14.02	Optimized	Catalytic rate constant for O-demethylmetoprolol formation.
$k_{cat,CYP2D6}$, αHM, NM default	1/min	6.02	8.27	R: Optimized. S: Mautz 1995	Catalytic rate constant for α-hydroxymetoprolol formation in simulations that use the NM model default.
$k_{cat,CYP2D6}$, ODM, NM default	1/min	9.87	10.37	R: Optimized. S: Mautz 1995	Catalytic rate constant for O-demethylmetoprolol formation in simulations that use the NM model default.
$CL_{spec,CYP3A4}$	1/min	0.022	0.021	Rüdesheim 2025	CYP3A4 first-order specific clearance. It replaces the unspecific hepatic clearance of the 2020 model.
GFR fraction	-	1.00	1.00	Assumed	Fraction used to scale passive glomerular filtration.
EHC fraction	-	1.00	1.00	Assumed	Fraction released continuously into bile.
P_{int}	cm/min	4.14×10^{-5}	4.14×10^{-5}	Optimized	Specific transcellular intestinal permeability.

Parameter	Unit	(R)-metoprolol	(S)-metoprolol	Source	Description
Partition coefficients	-	Rodgers and Rowland	Rodgers and Rowland	Calculated	Tissue-to-plasma partition coefficients calculated with the Rodgers and Rowland method.
Cellular permeabilities	cm/min	4.64×10^{-3}	4.64×10^{-3}	Calculated	Cellular permeabilities calculated with the PK-Sim Standard method.
$t_{50,IR}$	min	12.31	12.31	Optimized	Time to dissolve 50% of the immediate-release dose.
Weibull shape, IR	-	0.72	0.72	Optimized	Shape parameter for immediate-release dissolution.
$t_{50,CR}$	min	331.92	331.92	Optimized	Time to dissolve 50% of the controlled-release dose.
Weibull shape, CR	-	1.53	1.53	Optimized	Shape parameter for controlled-release dissolution.

Table 1: Drug-dependent parameters for (R)-metoprolol and (S)-metoprolol from Table 2, Table 3 and Supplement Table S2.3.2 of Rüdesheim 2020 and from Supplement Table S12 of Rüdesheim 2025. AS: activity score; CR: controlled release; EHC: enterohepatic circulation; GFR: glomerular filtration rate; IR: immediate release; ODM: O-demethylmetoprolol; αHM: α-hydroxymetoprolol.

AS	(R)-metoprolol k_{cat} → αHM [1/min]	(R)-metoprolol k_{cat} → ODM [1/min]	(S)-metoprolol k_{cat} → αHM [1/min]	(S)-metoprolol k_{cat} → ODM [1/min]	Origin
0	0.00	0.00	0.00	0.00	Assumed
0.5	1.65	2.70	1.82	2.27	Optimized
1.25	5.73	9.40	6.30	7.89	Optimized
1.5	6.38	10.48	7.03	8.81	Optimized
2	10.17	16.69	11.19	14.02	Optimized
3	19.03	31.22	20.93	26.23	Optimized

Table 1a: CYP2D6 activity-score-specific k_{cat} values for metoprolol metabolism from Table 3 and Supplement Table S3.1.1 of Rüdesheim 2020. AS = 0 represents the assumed absence of CYP2D6 activity. The other values were optimized. AS: activity score; ODM: O-demethylmetoprolol; αHM: α-hydroxymetoprolol.

Clinical data

The evaluation uses 47 plasma concentration-time profiles from adults (Table 2). Eighteen profiles were used for model building and 29 profiles were used for model verification. The assignments follow the training and test classifications in the published model. Seven verification profiles are the metoprolol-only control arms of the CYP2D6 interaction network (Rüdesheim 2025): Bennett 1982, Chellingsworth 1988 (two arms), Hemeryck 2000, Kirch 1982, Leemann 1993 (NM arm) and Toon 1988. Table S2.2.1 of Rüdesheim 2020 lists four further profiles that this evaluation does not include: Houtzagers 1982, Jack 1982, Krauwinkel 2013 and the Kelly 1985 oral single dose.

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Bae 2014	100	NR	NR	Male	3	Tablet IR	AS = 0.5 (IM)
Bae 2014	100	NR	NR	Male	4	Tablet IR	AS = 2 (NM)
Bennett 1982	100	28 (21-35)	71 (62-82)	Male	12	NR	EM
Chellingsworth 1988	100	22-34	NR	50% female	12	Tablet IR	EM
Chellingsworth 1988	100 b.i.d., multiple dose	23-32	NR	Male	12	Tablet IR	EM
Damy 2004+	200 q.d.	21-45	NR	27% female	13	Tablet CR	NM
Godbillon 1985	88.7, intravenous infusion over 150 min	23-29	NR	17% female	6	Solution	EM
Hamelin 2000	100	26 (20-36)	73 (59-96)	Male	10	Tablet IR	NM
Hamelin 2000	100	26 (20-36)	73 (59-96)	Male	6	Tablet IR	PM
Hemeryck 2000	100	20-29	NR	Male	8	Tablet IR	NM‡
Huang 1999+	100	24	63	Male	12	Tablet IR	AS = 0.5 (IM)
Huang 1999+	100	24	65	Male	12	Tablet IR	AS = 1.25 (NM)
Huang 1999+	100	25	65	Male	16	Tablet IR	AS = 2 (NM)
Jin 2008+	100	23	67	22% female	5	Tablet IR	AS = 0.5 (IM)
Jin 2008+	100	23	67	22% female	7	Tablet IR	AS = 1.25 (NM)
Jin 2008+	100	23	67	22% female	6	Tablet IR	AS = 2 (NM)

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Johnson and Burlew 1996+	200	29 (24-36)	82	Male	10	Tablet IR	NM
Johnson and Burlew 1996+	200	29 (24-40)	85	Male	10	Tablet IR	NM
Johnsson 1975	5, intravenous infusion over 10 min	23-28	62-70	Male	5	Solution	EM
Johnsson 1975	10, intravenous infusion over 10 min	23-28	62-70	Male	5	Solution	EM
Johnsson 1975	15, intravenous infusion over 10 min	23-28	62-70	Male	5	Solution	EM
Johnsson 1975	20, intravenous infusion over 10 min	23-28	62-70	Male	5	Solution	EM
Johnsson 1975	20	23-28	62-70	Male	5	Tablet IR	EM
Johnsson 1975	50	23-28	62-70	Male	5	Tablet IR	EM
Johnsson 1975	100	23-28	62-70	Male	5	Tablet IR	EM
Kelly 1985+	50, intravenous infusion over 10 min	19-26	60-98	Male	12	Solution	EM
Kelly 1985+	100 b.i.d., multiple dose	19-26	60-98	Male	12	NR	EM
Kirch 1982	100 b.i.d., multiple dose	24	69	Male	6	NR	EM
Kirchheiner 2004 / Seeringer 2008+	100	28 (23-34)	77 (69-81)	Male	13	Tablet IR	AS = 2 (NM)
Kirchheiner 2004 / Seeringer 2008+	100	38 (29-40)	90 (77-101)	Male	4	Tablet IR	AS = 0 (PM)
Kirchheiner 2004 / Seeringer 2008+	100	28 (25-37)	76 (70-80)	8% female	12	Tablet IR	AS = 3 (UM)
Krösser 2006	100	29 (18-39)	79 (62-100)	Male	18	Tablet IR	NM
Leemann 1993	20, intravenous infusion over 20 min	30	NR	Male	4	Solution	NM

Source	Dose [mg] / schedule*	Age [years]	Weight [kg]	Sex	N	Form.	CYP2D6 characterization
Luzier 1999	100 b.i.d., multiple dose	25 (21-35)	62 (54-77)	Female	10	Tablet IR	NM
Luzier 1999	100 b.i.d., multiple dose	26 (20-36)	84 (66-97)	Male	10	Tablet IR	NM
Parker and Soberman 2011	100	21-45	NR	27% female	15	Tablet CR	NM (model default)
Parker and Soberman 2011	100 b.i.d., multiple dose	21-45	NR	27% female	15	Tablet IR	NM (model default)
Parker and Soberman 2011	200	21-45	NR	27% female	15	Tablet CR	NM (model default)
Regårdh 1974[+]	5, intravenous infusion over 10 min	23-28	62-70	Male	5	Solution	EM
Regårdh 1974[+]	5	23-28	62-70	Male	5	Oral solution	EM
Regårdh and Johnsson 1980	10, intravenous infusion over 5 min	23-28	NR	Male	6	Solution	EM
Sharma 2005[+]	100	27 (18-40)	60 (49-100)	Female	16	NR	AS = 1.5 (NM)
Sharma 2005[+]	100	27 (18-40)	60 (49-100)	Female	4	NR	AS = 0 (PM)
Stout 2011	50	28 (18-45)	82 (63-94)	Male	10	NR	EM
Stout 2011	100	28 (18-45)	82 (63-94)	Male	10	Tablet CR	EM
Toon 1988	100 b.i.d., multiple dose	25	71	Male	12	NR	EM
Werner 2003	50	31	78	Male	12	Tablet IR	AS = 1.5 (NM)

Table 2: Clinical metoprolol concentration-time profiles used for model building and verification. *: Single oral dose unless otherwise specified; AS: activity score; b.i.d.: twice daily; CR: controlled release; EM: extensive metabolizer; IM: intermediate metabolizer; IR: immediate release; NM: normal metabolizer; NR: not reported; PM: poor metabolizer; PT: predicted phenotype; q.d.: once daily; UM: ultrarapid metabolizer; [+]: data used for model building; [‡]: simulated with the AS = 2 k_{cat} values, as in the paroxetine-metoprolol interaction model (Rüdesheim 2025). Parenthetical PTs for AS-coded rows

use the current CPIC CYP2D6 activity score-to-phenotype mapping (Moore 2026). EM is the model default when study-specific CYP2D6 information is not available. Only metoprolol-only study arms are included. Coadministered perpetrator arms are outside the scope of this compound report.

2.3 Model parameters and assumptions

Absorption

The model includes oral administration of metoprolol tartrate as a normal-release formulation, metoprolol succinate as a controlled-release formulation and dissolved oral dosing. Formulation behavior is represented through Weibull dissolution parameters for normal-release and controlled-release products, with the same compound intestinal permeability applied to both metoprolol enantiomers.

Oral absorption was evaluated together with first-pass metabolism because most clinical studies report plasma concentration-time data after oral dosing. Intravenous infusion studies were retained to separate systemic disposition from oral formulation behavior where possible.

The specific intestinal permeability was optimized and applied consistently for (R)- and (S)-metoprolol. Differences between CYP2D6 activity-score groups are assigned to metabolism rather than to absorption.

Distribution

(R)- and (S)-metoprolol use a fraction unbound of 88% and a logP of 1.77 as summarized in Section 2.2. The same distribution assumptions are used for both enantiomers because the source information does not support separate enantiomer-specific binding or lipophilicity values.

Partition coefficients were calculated with the Rodgers and Rowland method. Cellular permeabilities were calculated with the PK-Sim Standard method. The racemic metoprolol observer is used only for comparison with total metoprolol observations and does not replace the enantiomer-specific disposition model.

α-Hydroxymetoprolol is represented with its own molecular weight, lipophilicity, plasma binding and clearance parameters. This separation is required because metabolite exposure is controlled by both formation from the parent enantiomers and metabolite-specific elimination.

The α-hydroxymetoprolol solubility input is the HMDB ALOGPS prediction of 1.43 g/L, implemented as 1,430 mg/L (Wishart 2018). Table S2.4.3 of the published model labels this value as g/mL, but the cited HMDB source establishes g/L.

Metabolism and elimination

Metoprolol clearance is represented by CYP2D6-dependent α-hydroxylation, CYP2D6-dependent O-demethylation, CYP3A4 first-order clearance and renal filtration.

- CYP2D6

CYP2D6 is the dominant metabolic enzyme in the model. Separate CYP2D6 pathways are implemented for (R)- and (S)-metoprolol α-hydroxylation and O-demethylation. K_m values are

enantiomer- and pathway-specific, while k_{cat} values are scaled by CYP2D6 activity score as documented in Table 3 of Rüdesheim 2020.

The CYP2D6 activity-score implementation is the key determinant of simulated exposure differences between poor, intermediate, normal and ultrarapid metabolizer groups. Poor-metabolizer activity is set to zero, while non-zero activity-score groups use optimized k_{cat} values. Simulations without an activity-score group use the NM default k_{cat} values in Table 1. For (S)-metoprolol, these values equal the in vitro values of Mautz 1995, as listed in Supplement Table S12 of Rüdesheim 2025. They differ from the optimized values of 6.55 and 8.21 1/min in Rüdesheim 2020.

- CYP3A4

CYP3A4 is implemented as a first-order clearance pathway for each metoprolol enantiomer. This pathway accounts for CYP2D6-independent oxidative clearance and supports simulations in CYP2D6 poor-metabolizer groups without assigning all remaining clearance to renal elimination. This pathway replaces the unspecific hepatic clearance process of the 2020 model (Rüdesheim 2025).

The CYP3A4 clearance is empirical. The fraction of metoprolol metabolized by CYP3A4 in vivo is not known (Rüdesheim 2020).

- Renal and metabolite elimination

Renal filtration is included with a GFR fraction of 1 for metoprolol and α-hydroxymetoprolol. α-Hydroxymetoprolol also includes unspecific hepatic clearance to describe metabolite elimination beyond passive filtration.

3 Results and Discussion

The evaluation combines goodness-of-fit diagnostics and concentration-time profiles for racemic metoprolol, (R)-metoprolol, (S)-metoprolol and α-hydroxymetoprolol. The diagnostic plots are separated by analyte to avoid ambiguous legends.

The intravenous studies assess systemic disposition. The oral immediate-release tartrate and controlled-release succinate studies assess formulation and first-pass behavior. The CYP2D6-stratified studies assess the activity-score-dependent metabolic implementation.

The evaluation includes adult metoprolol-only study arms. It does not evaluate coadministered perpetrator arms. Residual uncertainty includes genotype-level differences within the same CYP2D6 activity score and the quantitative contribution of non-CYP2D6 metabolism.

3.1 Metoprolol final input parameters

The following tables summarize the final drug-dependent model input parameters generated from the model. Parameter values should be interpreted together with the source and method information in Section 2.2 and Section 2.3.

Compound: R-Metoprolol

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1000 mg/ml	Publication-Avdeef 2001	metoprolol tartrate	True
Reference pH	7.4	Unknown	metoprolol tartrate	True
Solubility at reference pH	157 mg/ml	Publication-Boldhane 2010	metoprolol succinate	False
Reference pH	5.5	Publication-Boldhane 2010	metoprolol succinate	False
Lipophilicity	1.77 Log Units	Publication-Other-Zhao 2002	logP	True
Fraction unbound (plasma, reference value)	0.88	Publication-Benfield 1986	fu	True
Specific intestinal permeability (transcellular)	4.1375817791E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Is small molecule	Yes			
Molecular weight	267.36 g/mol	Publication-Kim 2019		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-ASSUMPTION

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Unknown

Metabolizing Enzyme: CYP3A4-Assumption

Species: Human

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Intrinsic clearance	0 l/min	
Specific clearance	0.0219977664 1/min	Publication-Rüdesheim 2025, Table S12

Metabolizing Enzyme: CYP2D6-NM, Mautz 1995 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0.123 nmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.82 µmol/l	Publication-In Vitro-Mautz 1995
kcat	9.8720570601 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-NM, Mautz 1995 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0.075 nmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.08 µmol/l	Publication-In Vitro-Mautz 1995
kcat	6.0195469879 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=2 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.82 µmol/l	Publication-In Vitro-Mautz 1995
kcat	16.6917426582 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=2 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.08 µmol/l	Publication-In Vitro-Mautz 1995
kcat	10.1731346705 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.25 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.08 µmol/l	Publication-In Vitro-Mautz 1995
kcat	5.7303081004 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.25 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.82 µmol/l	Publication-In Vitro-Mautz 1995
kcat	9.3972567283 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.5 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.08 µmol/l	Publication-In Vitro-Mautz 1995
kcat	6.3836485827 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.5 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.82 µmol/l	Publication-In Vitro-Mautz 1995
kcat	10.4784549353 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=0.5 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.82 µmol/l	Publication-In Vitro-Mautz 1995
kcat	2.703131627 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=0.5 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.08 µmol/l	Publication-In Vitro-Mautz 1995
kcat	1.6472208352 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-PM, AS=0 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.82 µmol/l	Publication-In Vitro-Mautz 1995

Metabolizing Enzyme: CYP2D6-PM, AS=0 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.08 µmol/l	Publication-In Vitro-Mautz 1995

Metabolizing Enzyme: CYP2D6-AS=3 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	8.82 µmol/l	Publication-In Vitro-Mautz 1995
kcat	31.22 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=3 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.08 µmol/l	Publication-In Vitro-Mautz 1995
kcat	19.03 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: S-Metoprolol

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1000 mg/ml	Publication-Avdeef 2001	metoprolol tartrate	True
Reference pH	7.4	Unknown	metoprolol tartrate	True
Solubility at reference pH	157 mg/ml	Publication-Boldhane 2010	metoprolol succinate	False
Reference pH	5.5	Publication-Boldhane 2010	metoprolol succinate	False
Lipophilicity	1.77 Log Units	Publication-Other-Zhao 2002	logP	True
Fraction unbound (plasma, reference value)	0.88	Publication-Benfield 1986	fu	True
Specific intestinal permeability (transcellular)	4.1375817791E-05 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
Is small molecule	Yes			
Molecular weight	267.36 g/mol	Publication-Kim 2019		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-Paper

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Unknown

Metabolizing Enzyme: CYP3A4-Assumption

Species: Human

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
Intrinsic clearance	0 l/min	
Specific clearance	0.0212570166 1/min	Publication-Rüdesheim 2025, Table S12

Metabolizing Enzyme: CYP2D6-NM, Mautz 1995 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0.1037 nmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	12.43 µmol/l	Publication-In Vitro-Mautz 1995

Metabolizing Enzyme: CYP2D6-NM, Mautz 1995 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0.0827 nmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.75 µmol/l	Publication-In Vitro-Mautz 1995

Metabolizing Enzyme: CYP2D6-AS=0 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.75 µmol/l	Publication-In Vitro-Mautz 1995

Metabolizing Enzyme: CYP2D6-AS=0 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	12.43 µmol/l	Publication-In Vitro-Mautz 1995

Metabolizing Enzyme: CYP2D6-AS=0.5 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	12.43 µmol/l	Publication-In Vitro-Mautz 1995
kcat	2.2702082024 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.25 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	12.43 µmol/l	Publication-In Vitro-Mautz 1995
kcat	7.8945927646 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.5 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	12.43 µmol/l	Publication-In Vitro-Mautz 1995
kcat	8.8076966829 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=2 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	12.43 µmol/l	Publication-In Vitro-Mautz 1995
kcat	14.0227063168 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=0.5 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.75 µmol/l	Publication-In Vitro-Mautz 1995
kcat	1.8161665619 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.25 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.75 µmol/l	Publication-In Vitro-Mautz 1995
kcat	6.3022175194 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.5 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.75 µmol/l	Publication-In Vitro-Mautz 1995
kcat	7.0307052469 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=2 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.75 µmol/l	Publication-In Vitro-Mautz 1995
kcat	11.19 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=3 (α-Hydroxymetoprolol)

Molecule: CYP2D6

Metabolite: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	10.75 µmol/l	Publication-In Vitro-Mautz 1995
kcat	20.93 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=3 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	12.43 µmol/l	Publication-In Vitro-Mautz 1995
kcat	26.23 1/min	Parameter Identification-Parameter Identification-Optimized

Compound: Metoprolol racemate

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	16900 mg/l	Publication-In Vitro-Avdeef 2001	metoprolol tartrate	True
Reference pH	7		metoprolol tartrate	True
Lipophilicity	1.77 Log Units	Publication-Other-Zhao 2002	logP	True
Fraction unbound (plasma, reference value)	0.88	Publication-Benfield 1986	fu	True
Specific intestinal permeability (transcellular)	4.1375817791E-05 cm/min		Optimized	True
Is small molecule	Yes			
Molecular weight	267.36 g/mol	Database-PubChem		
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Compound: alpha-Hydroxymetoprolol

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	1430 mg/l	Database-Other-HMDB 4.0 (ALOGPS predicted water solubility, 1.43 g/l)	Predicted	True
Reference pH	7	Database-Other-HMDB	Predicted	True
Lipophilicity	0.8658570657 Log Units	Parameter Identification-Parameter Identification-optimized	logP	True
Fraction unbound (plasma, reference value)	0.6251	Publication-Other-Watanabe 2018	Predicted	True
Is small molecule	Yes			
Molecular weight	283.36 g/mol			
Plasma protein binding partner	Albumin			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Glomerular Filtration-ASSUMPTION

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	Other-Assumption

Systemic Process: Total Hepatic Clearance-ASSUMPTION

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.12	
Lipophilicity (experiment)	0.47 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	0.34 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

The goodness-of-fit diagnostics are presented separately for racemic metoprolol, (R)-metoprolol, (S)-metoprolol, and α-hydroxymetoprolol. This separation prevents analyte-dependent prediction errors from being hidden in a combined statistic or an ambiguous legend.

3.2.1 Metoprolol racemate goodness-of-fit diagnostics

The racemic metoprolol GOF plot compares total metoprolol plasma observations with the racemic metoprolol model output.

Table 3-1: GMFE for Metoprolol racemate goodness-of-fit diagnostics

Group	GMFE
Metoprolol racemate	1.34

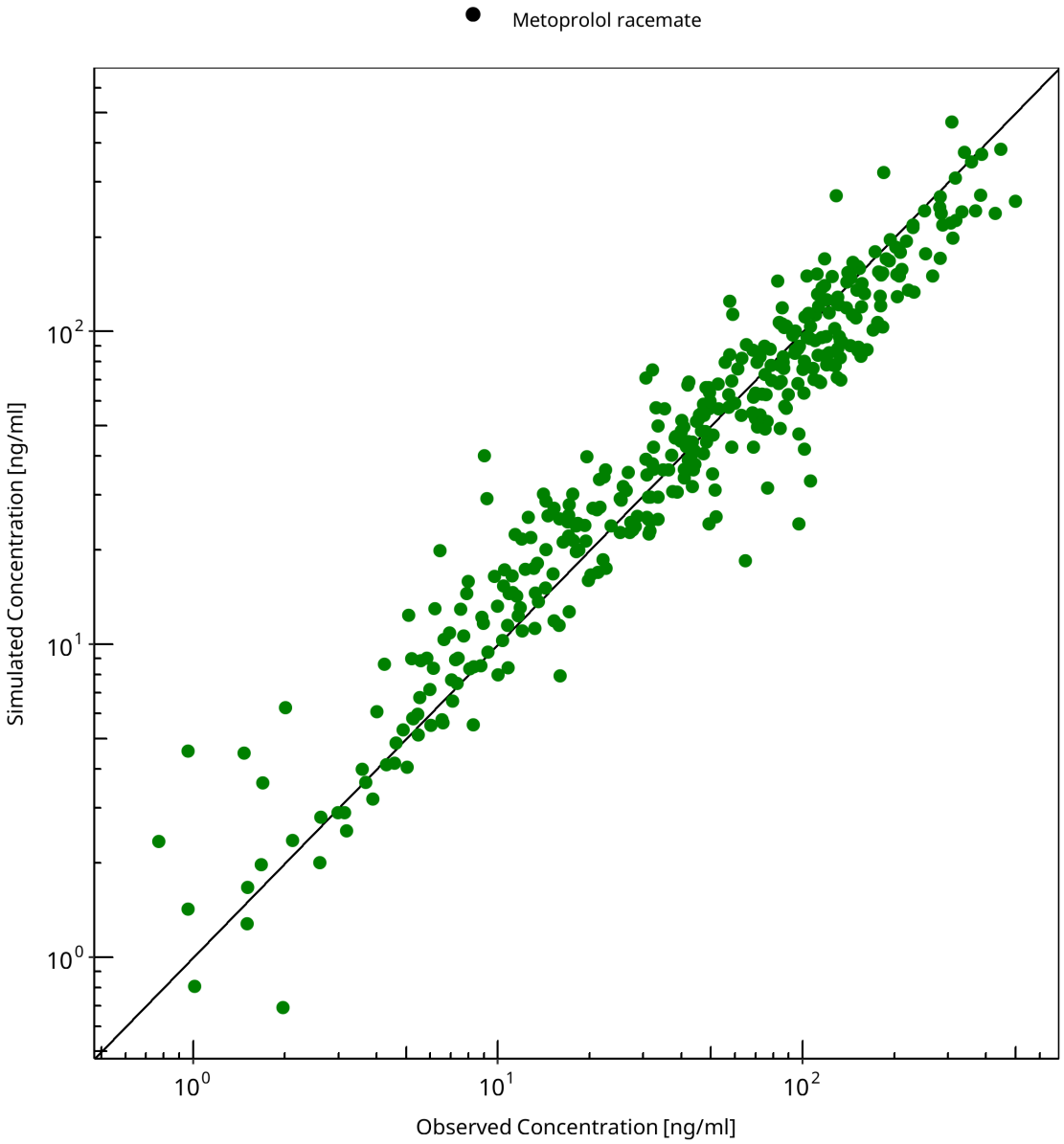


Figure 3-1: Metoprolol racemate goodness-of-fit diagnostics

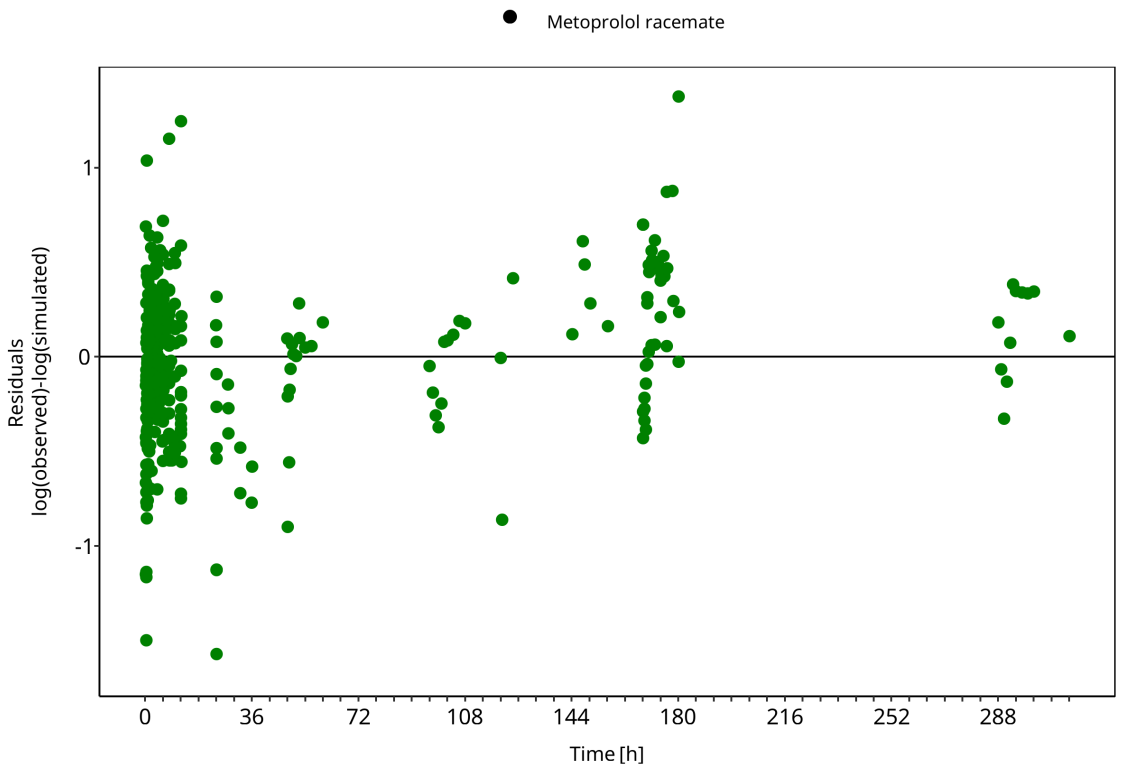


Figure 3-2: Metoprolol racemate goodness-of-fit diagnostics

3.2.2 (R)-Metoprolol goodness-of-fit diagnostics

The (R)-metoprolol GOF plot evaluates the enantiomer-specific parent concentration predictions across studies reporting (R)-metoprolol observations.

Table 3-2: GMFE for (R)-Metoprolol goodness-of-fit diagnostics

Group	GMFE
R-Metoprolol	1.29

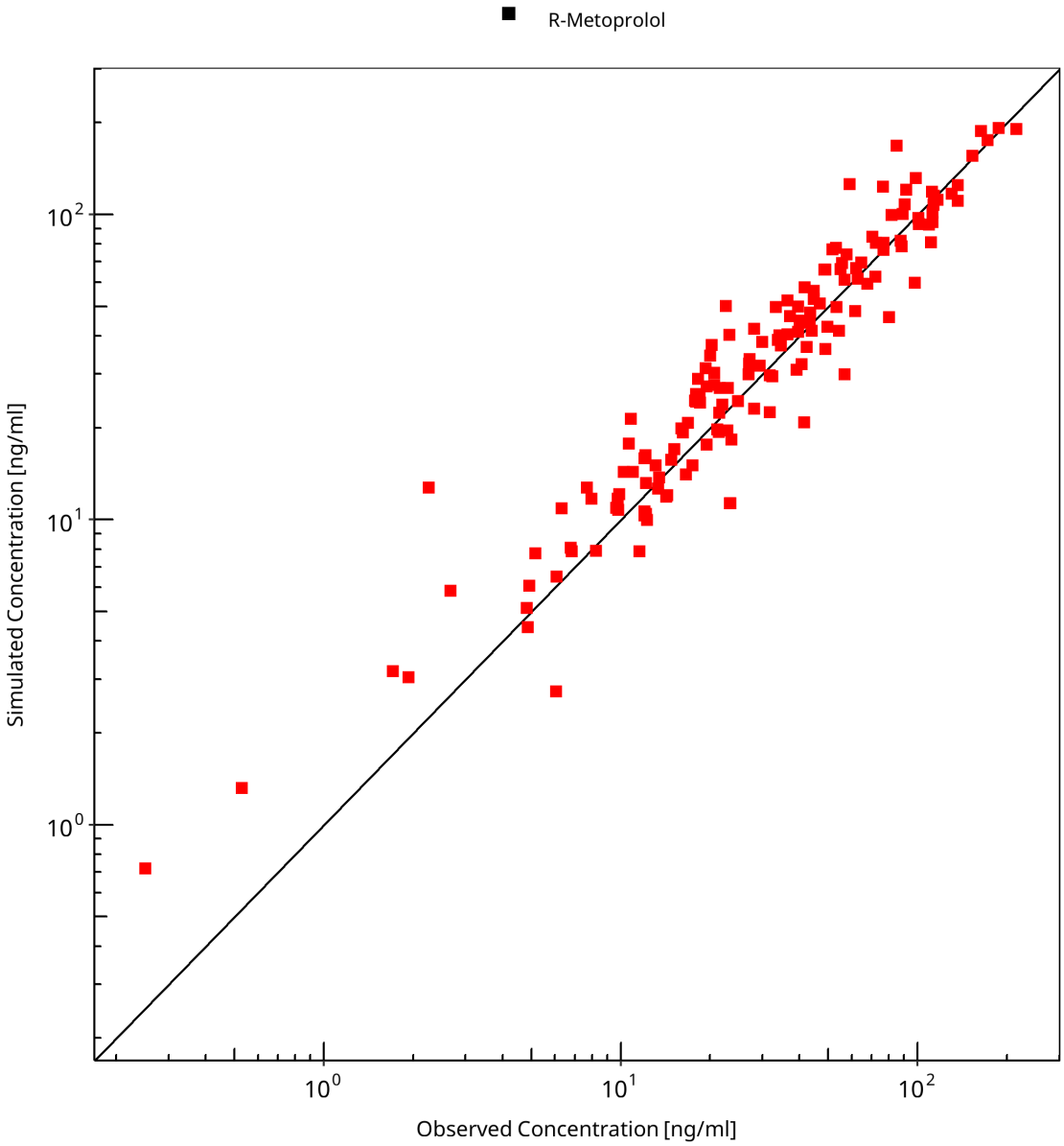


Figure 3-3: (R)-Metoprolol goodness-of-fit diagnostics

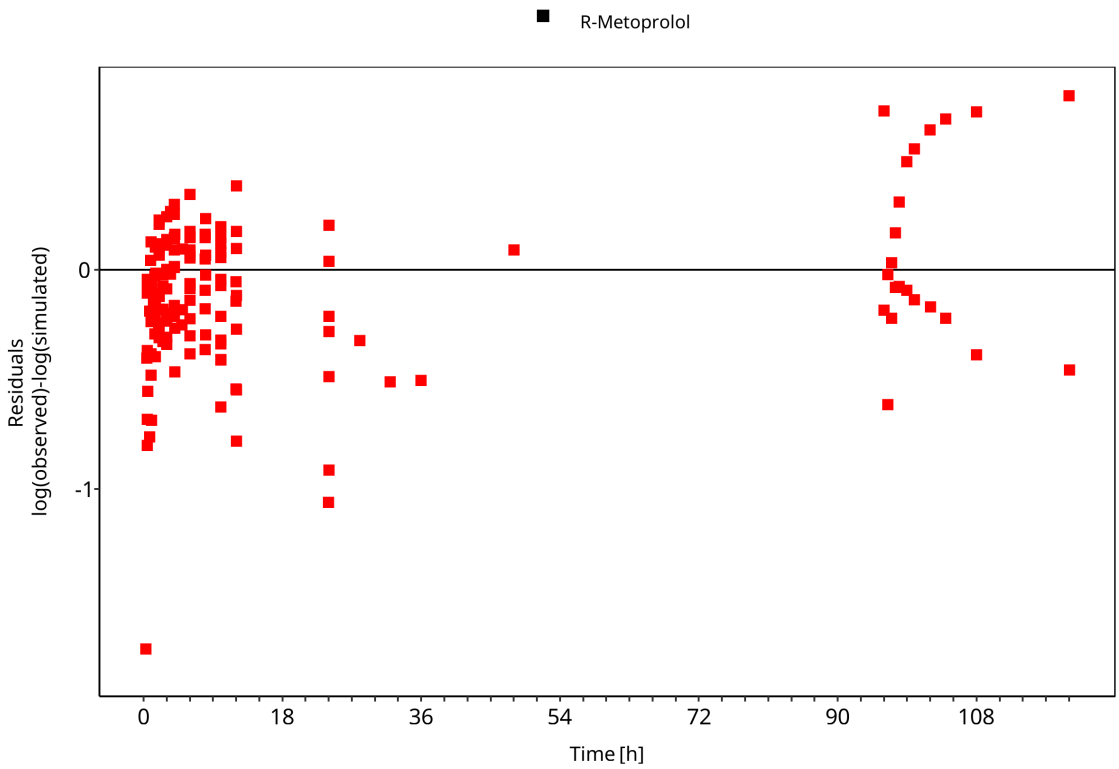


Figure 3-4: (R)-Metoprolol goodness-of-fit diagnostics

3.2.3 (S)-Metoprolol goodness-of-fit diagnostics

The (S)-metoprolol GOF plot evaluates the enantiomer-specific parent concentration predictions across studies reporting (S)-metoprolol observations.

Table 3-3: GMFE for (S)-Metoprolol goodness-of-fit diagnostics

Group	GMFE
S-Metoprolol	1.30

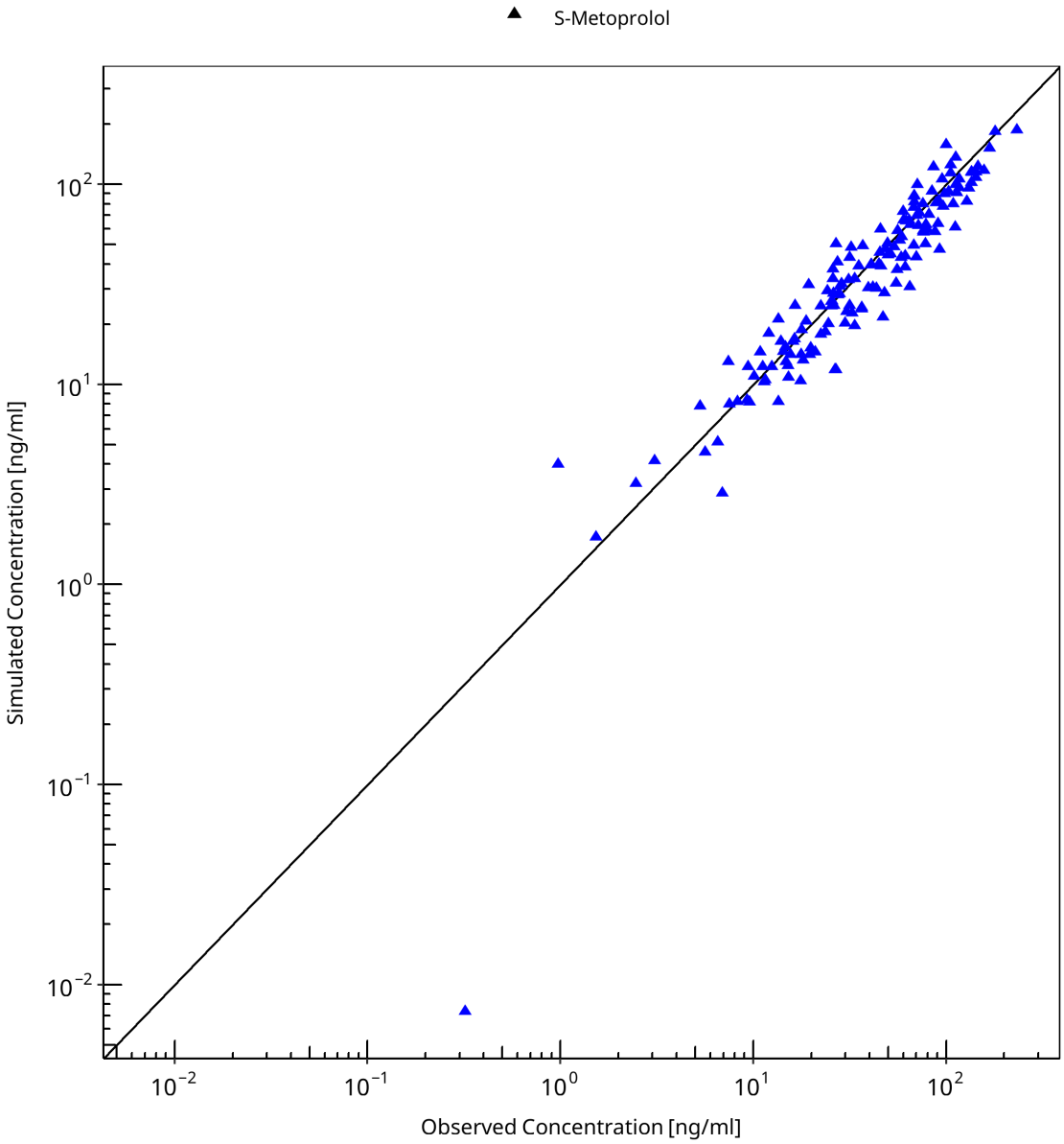


Figure 3-5: (S)-Metoprolol goodness-of-fit diagnostics

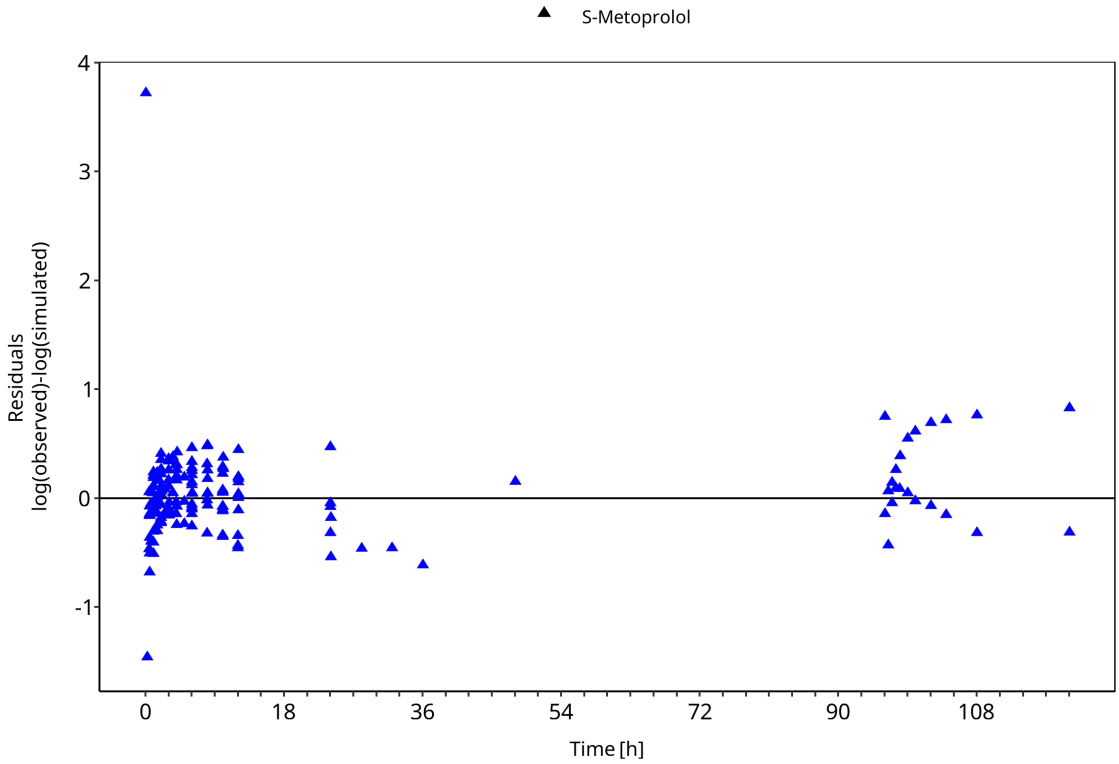


Figure 3-6: (S)-Metoprolol goodness-of-fit diagnostics

3.2.4 α-Hydroxymetoprolol goodness-of-fit diagnostics

The α-hydroxymetoprolol GOF plot evaluates the metabolite concentration predictions for studies reporting metabolite plasma concentrations.

Table 3-4: GMFE for α-Hydroxymetoprolol goodness-of-fit diagnostics

Group	GMFE
alpha-Hydroxymetoprolol	1.83

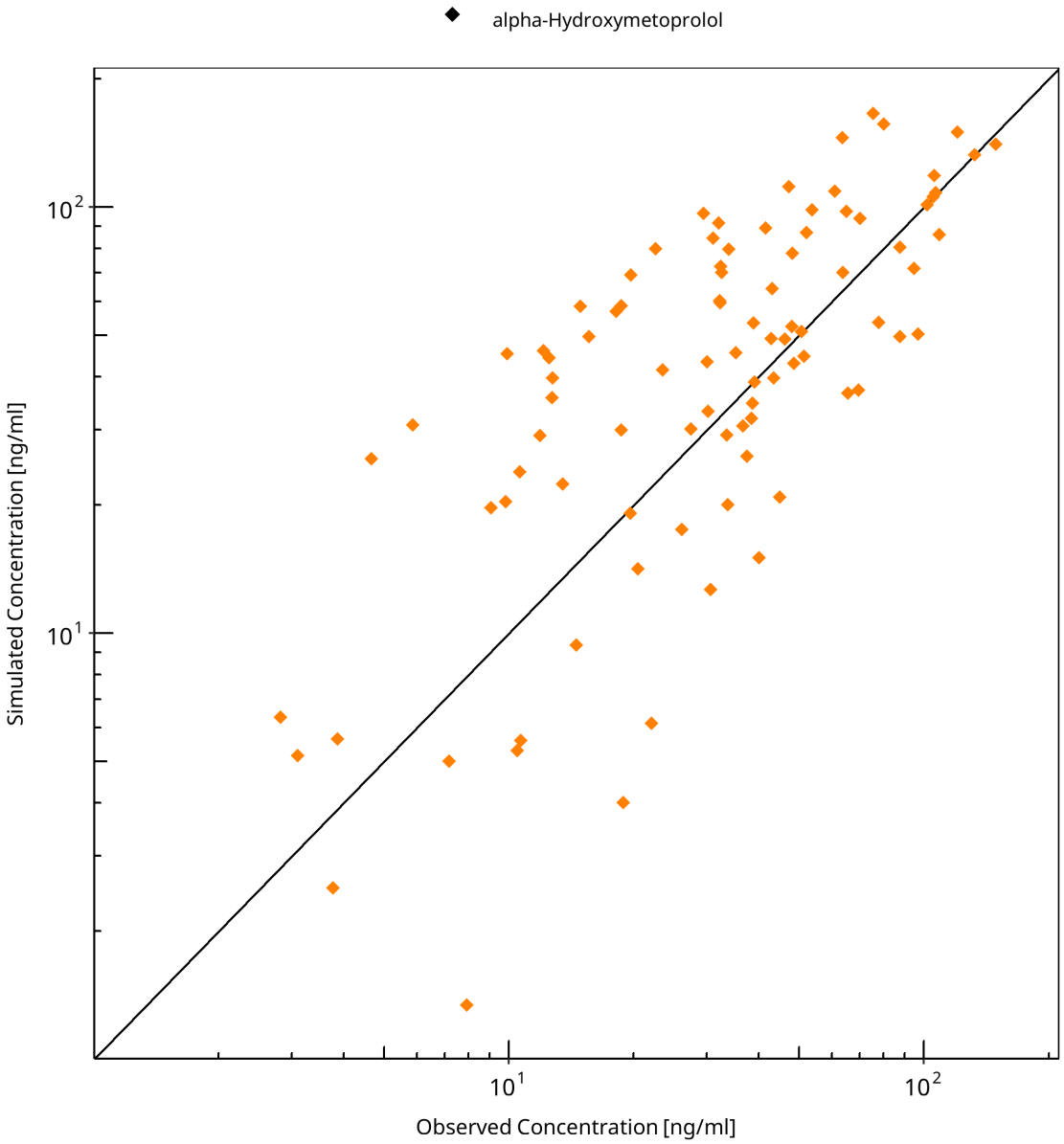


Figure 3-7: α-Hydroxymetoprolol goodness-of-fit diagnostics

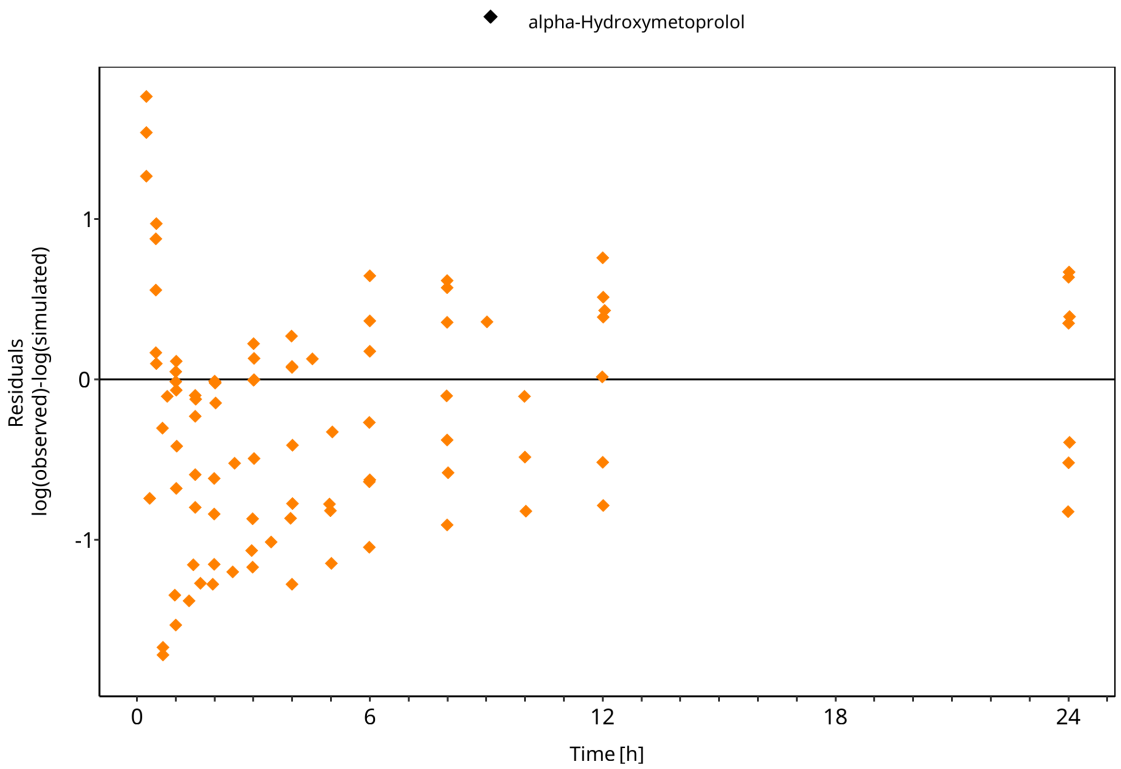


Figure 3-8: α-Hydroxymetoprolol goodness-of-fit diagnostics

3.3 Concentration-time profiles

The concentration-time profiles show simulated and observed plasma concentrations for the model-building and model-verification datasets. Profiles include intravenous infusion, oral normal-release metoprolol tartrate and oral controlled-release metoprolol succinate studies.

3.3.1 Model Building

Model-building profiles include studies used to inform systemic disposition, absorption, formulation behavior and CYP2D6 activity-score-dependent metabolism.

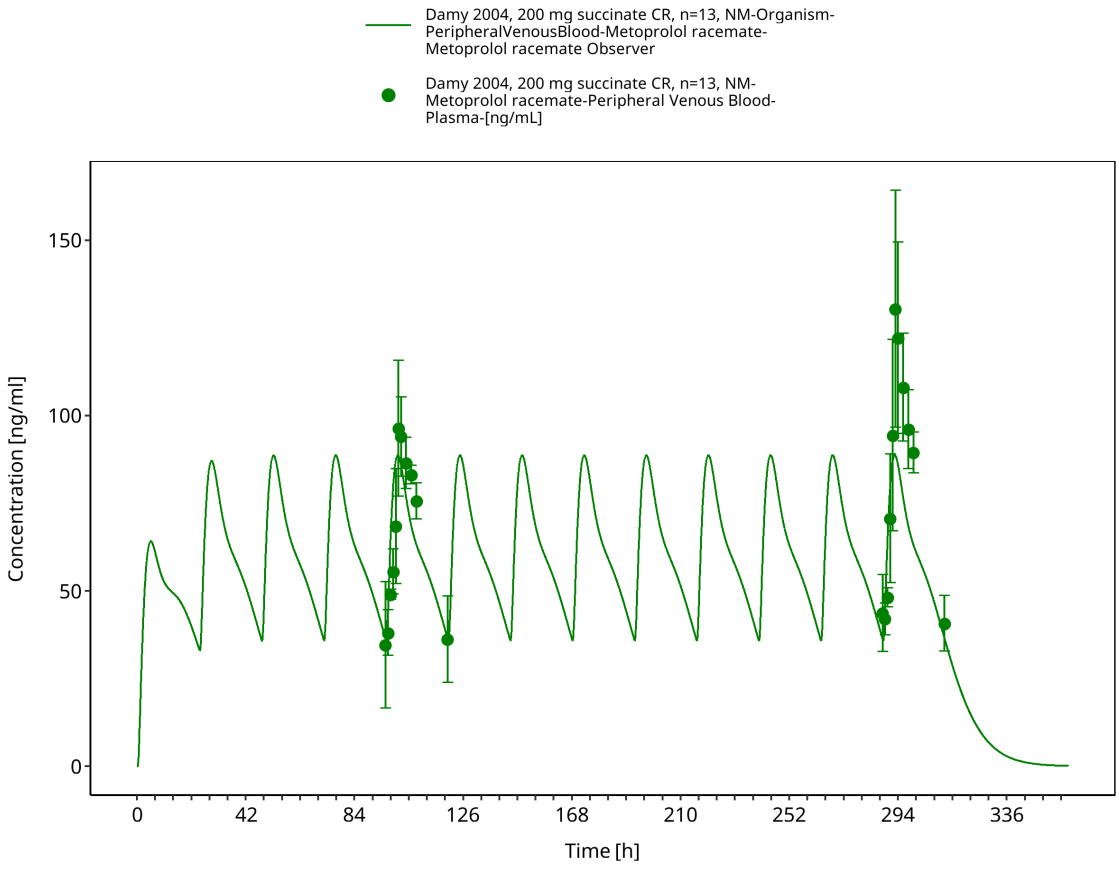


Figure 3-9: Damy 2004: 200 mg succinate CR, n=13, NM

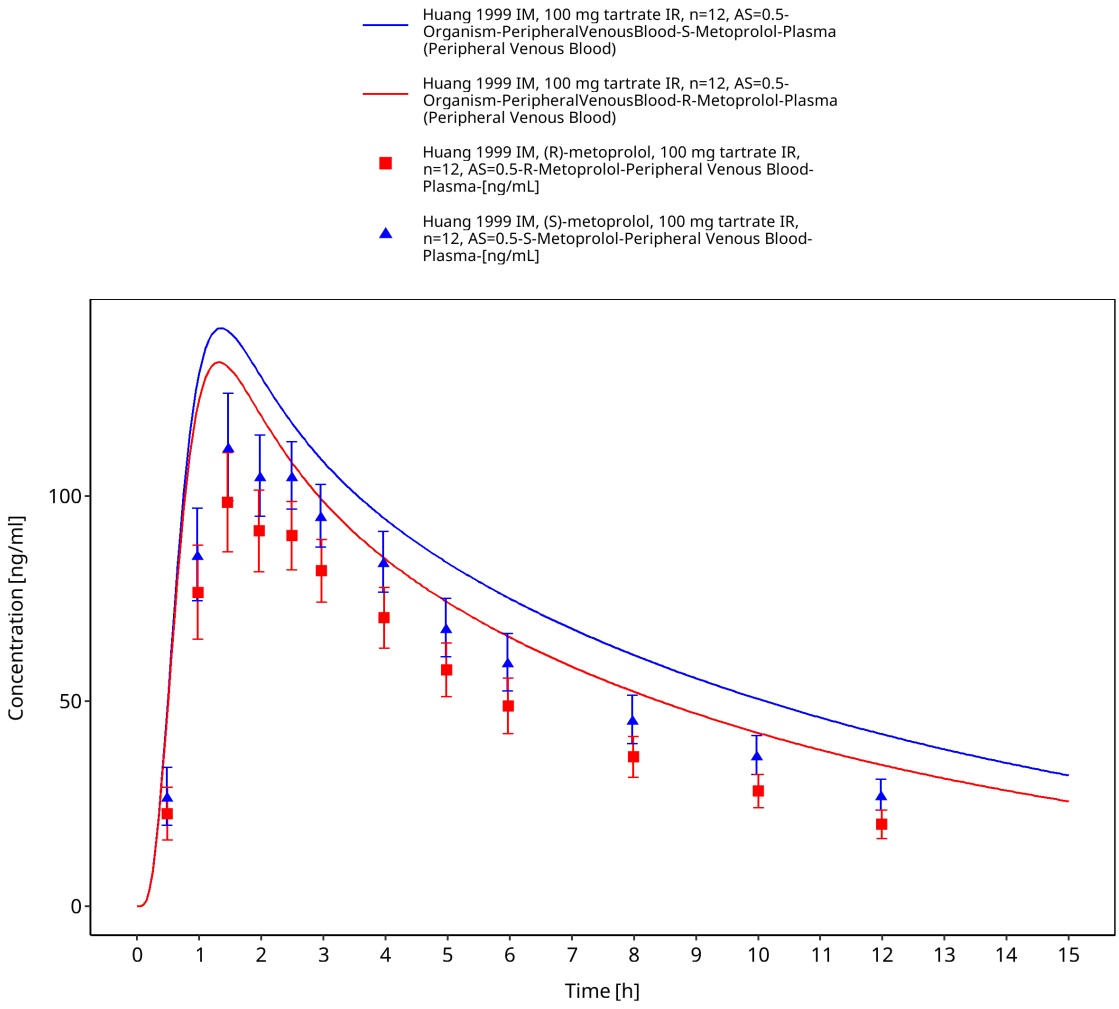


Figure 3-10: Huang 1999: IM, 100 mg tartrate IR, n=12, AS=0.5

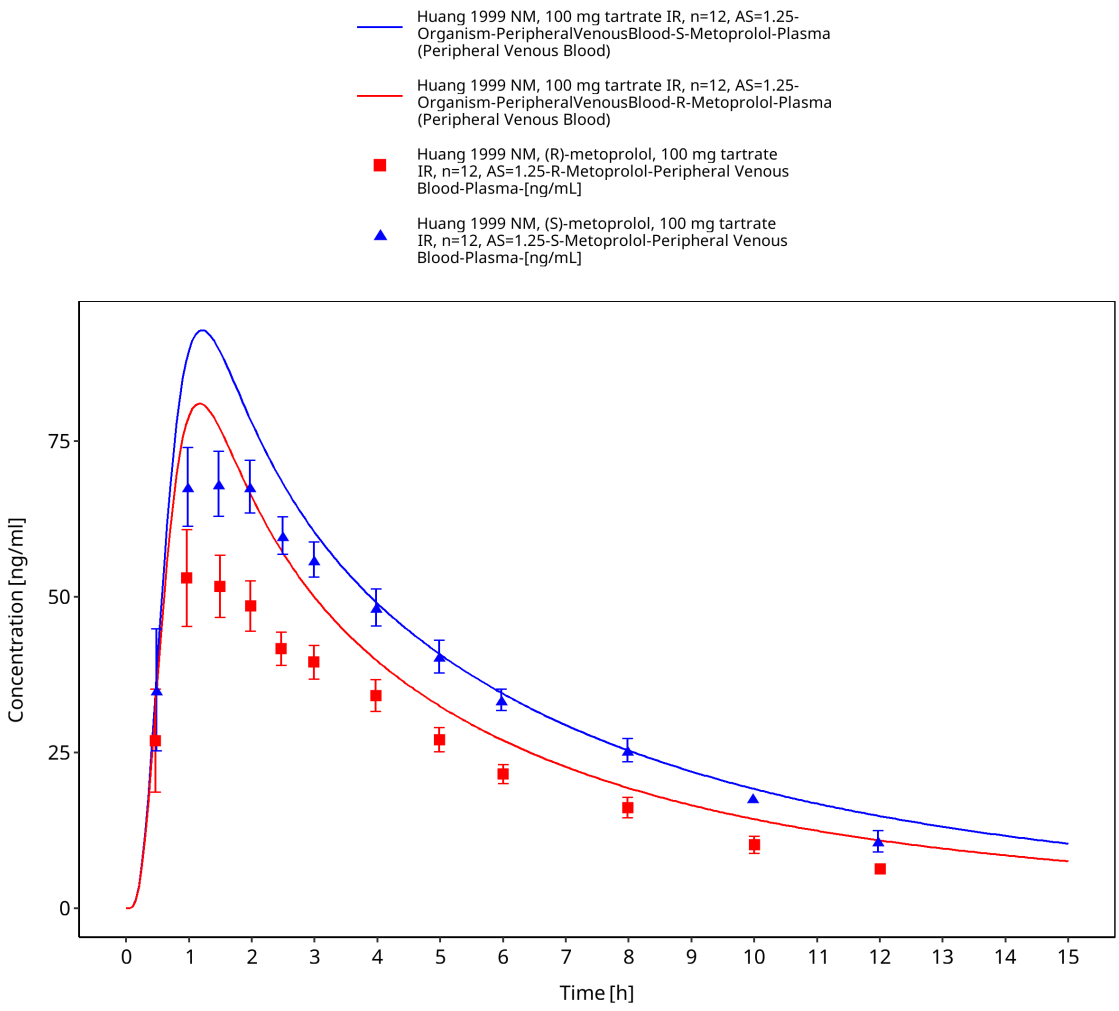


Figure 3-11: Huang 1999: NM, 100 mg tartrate IR, n=12, AS=1.25

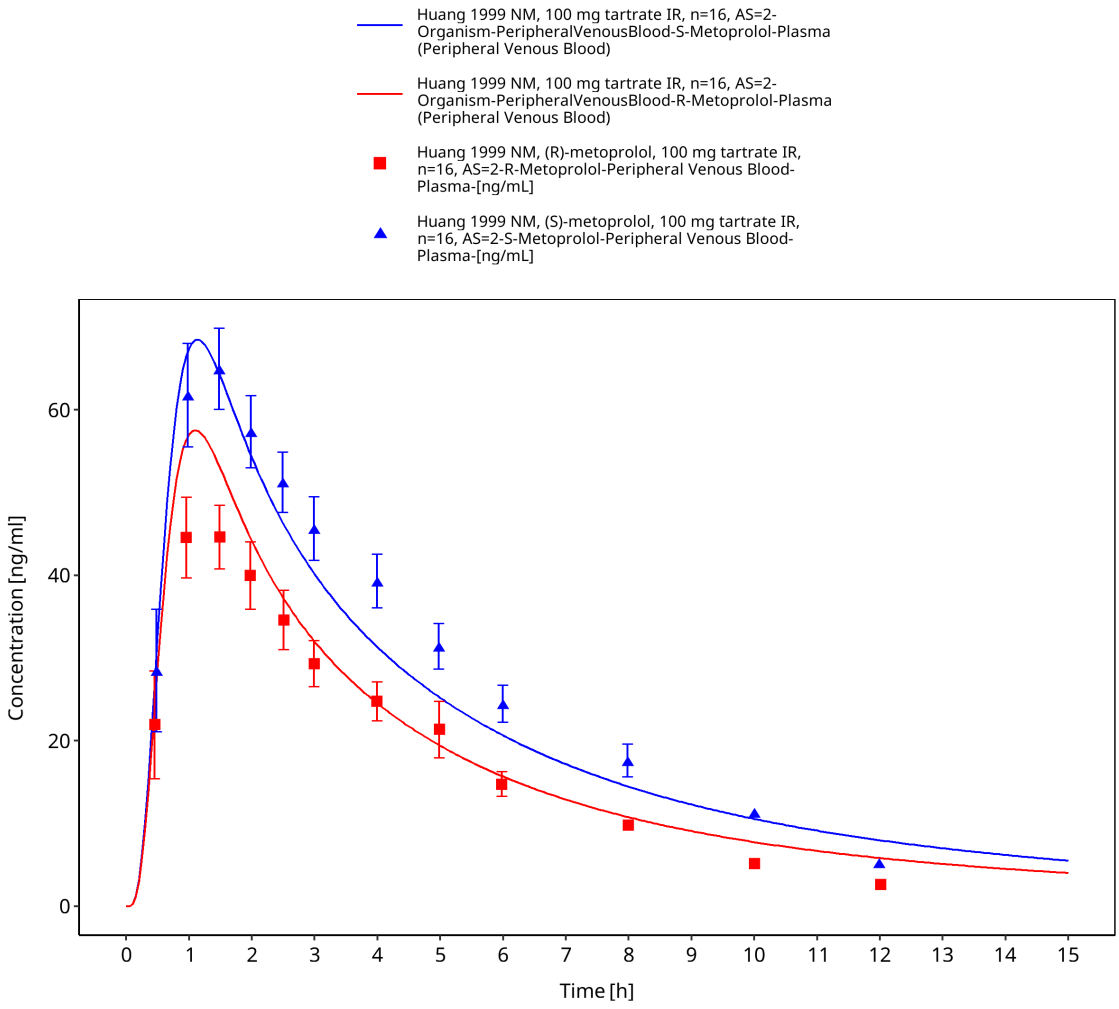


Figure 3-12: Huang 1999: NM, 100 mg tartrate IR, n=16, AS=2

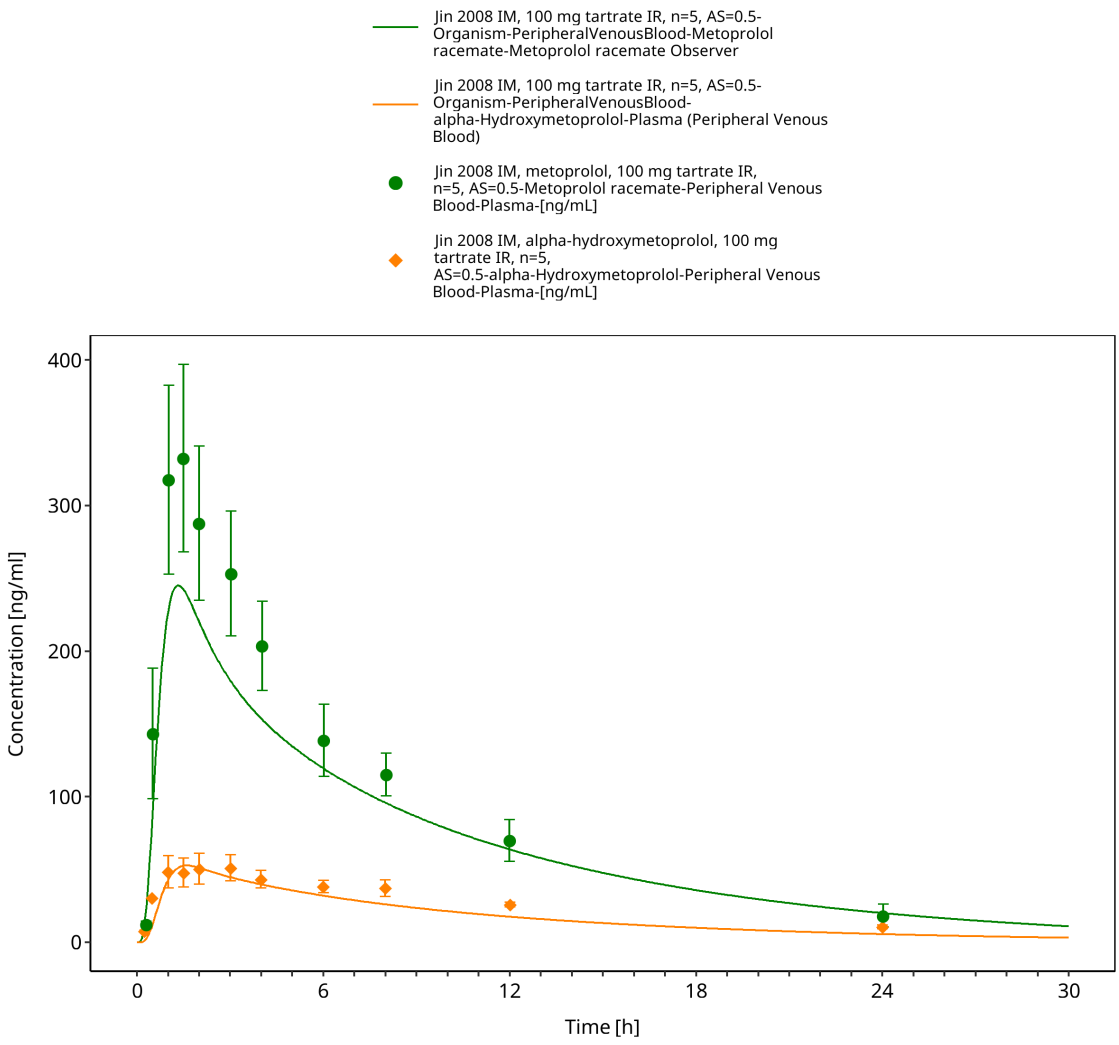


Figure 3-13: Jin 2008: IM, 100 mg tartrate IR, n=5, AS=0.5

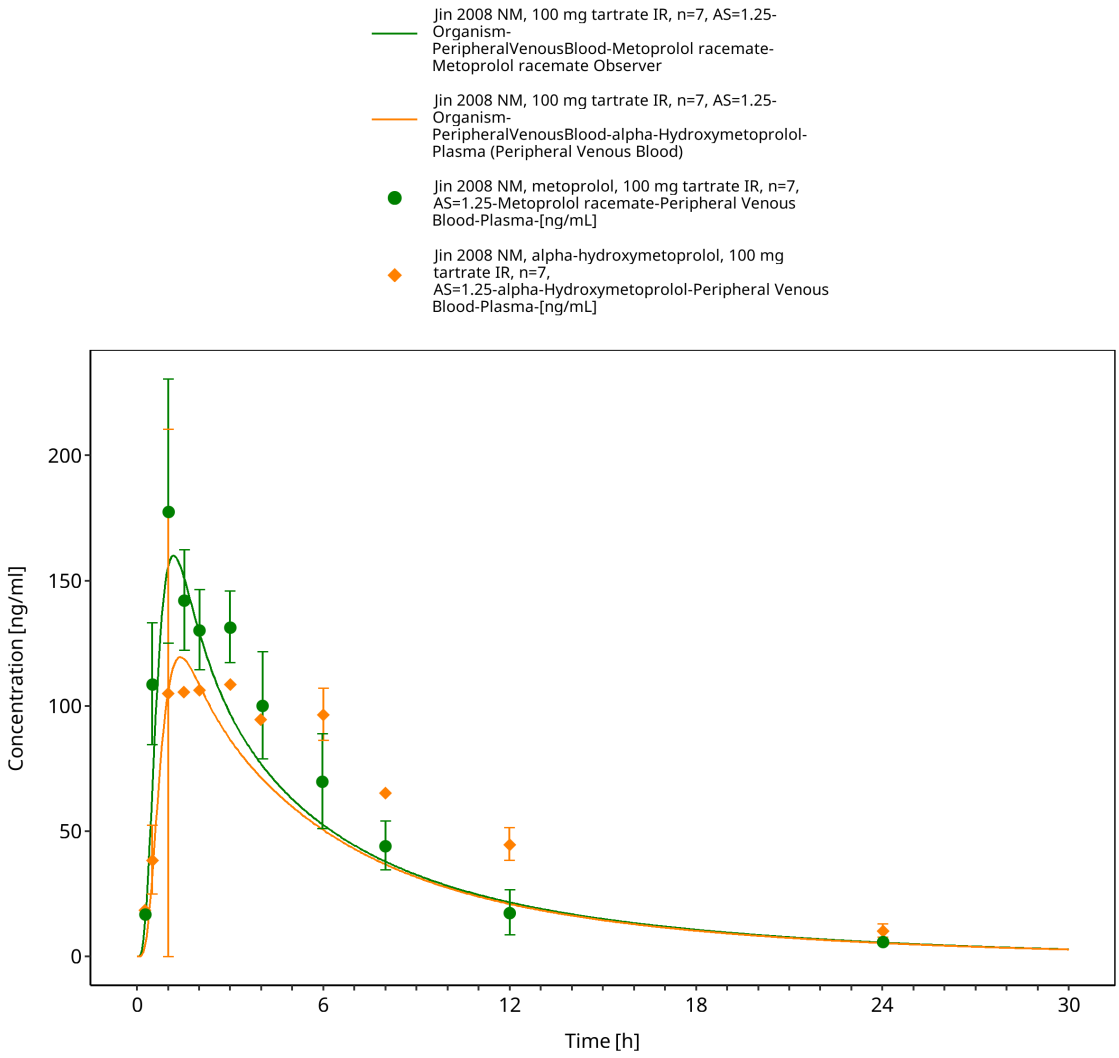


Figure 3-14: Jin 2008: NM, 100 mg tartrate IR, n=7, AS=1.25

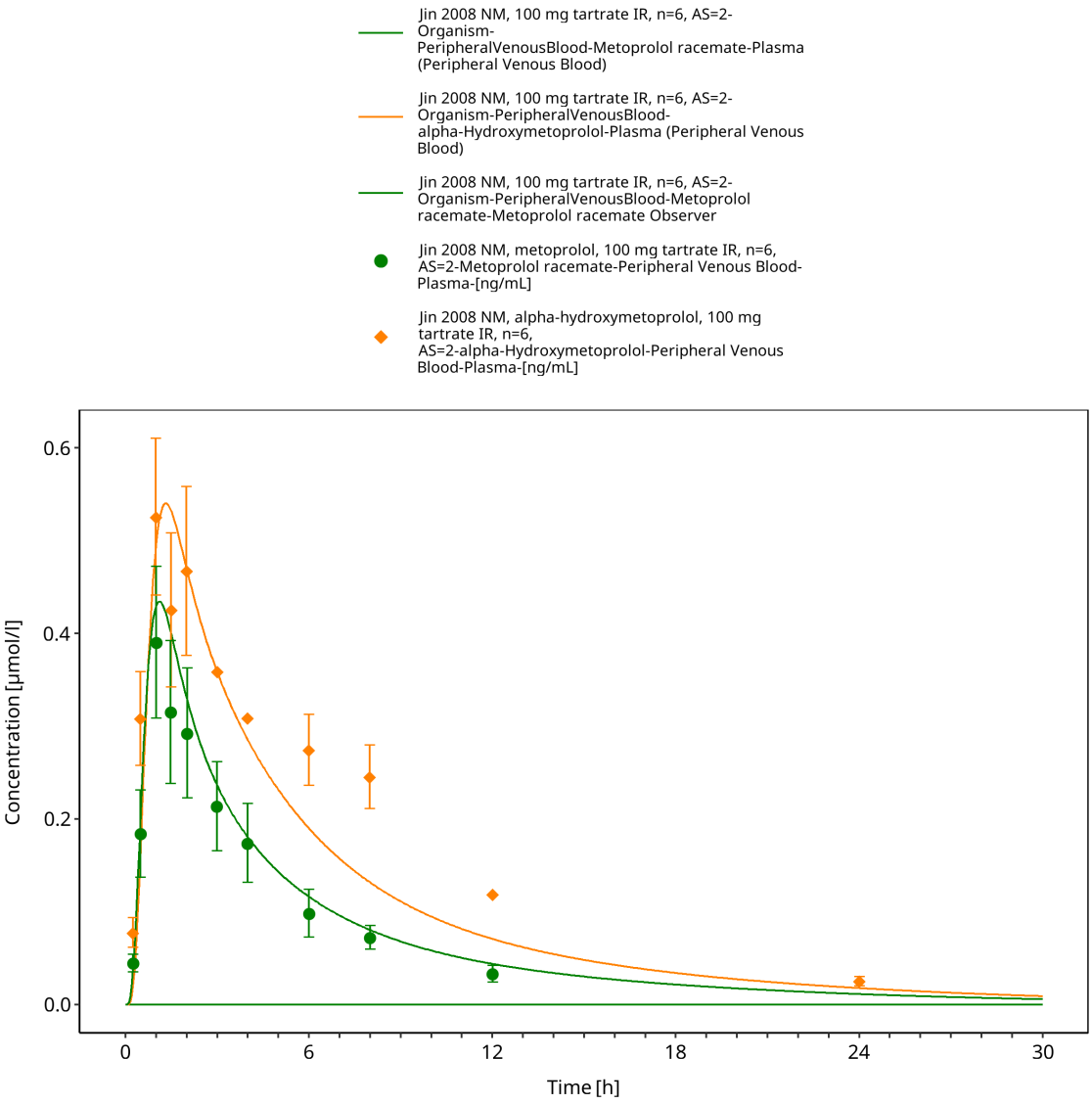


Figure 3-15: Jin 2008: NM, 100 mg tartrate IR, n=6, AS=2

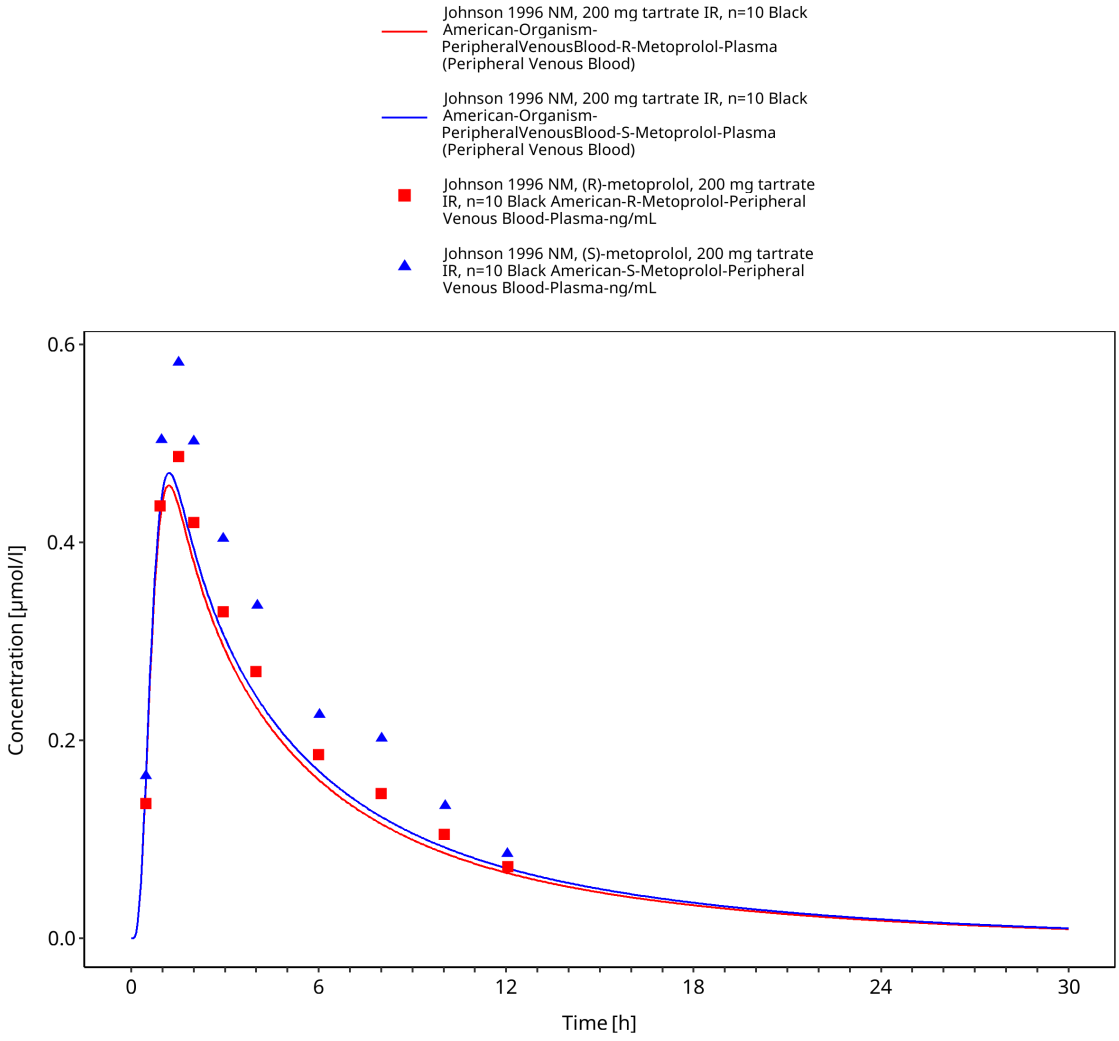


Figure 3-16: Johnson 1996: NM, 200 mg tartrate IR, n=10 Black American

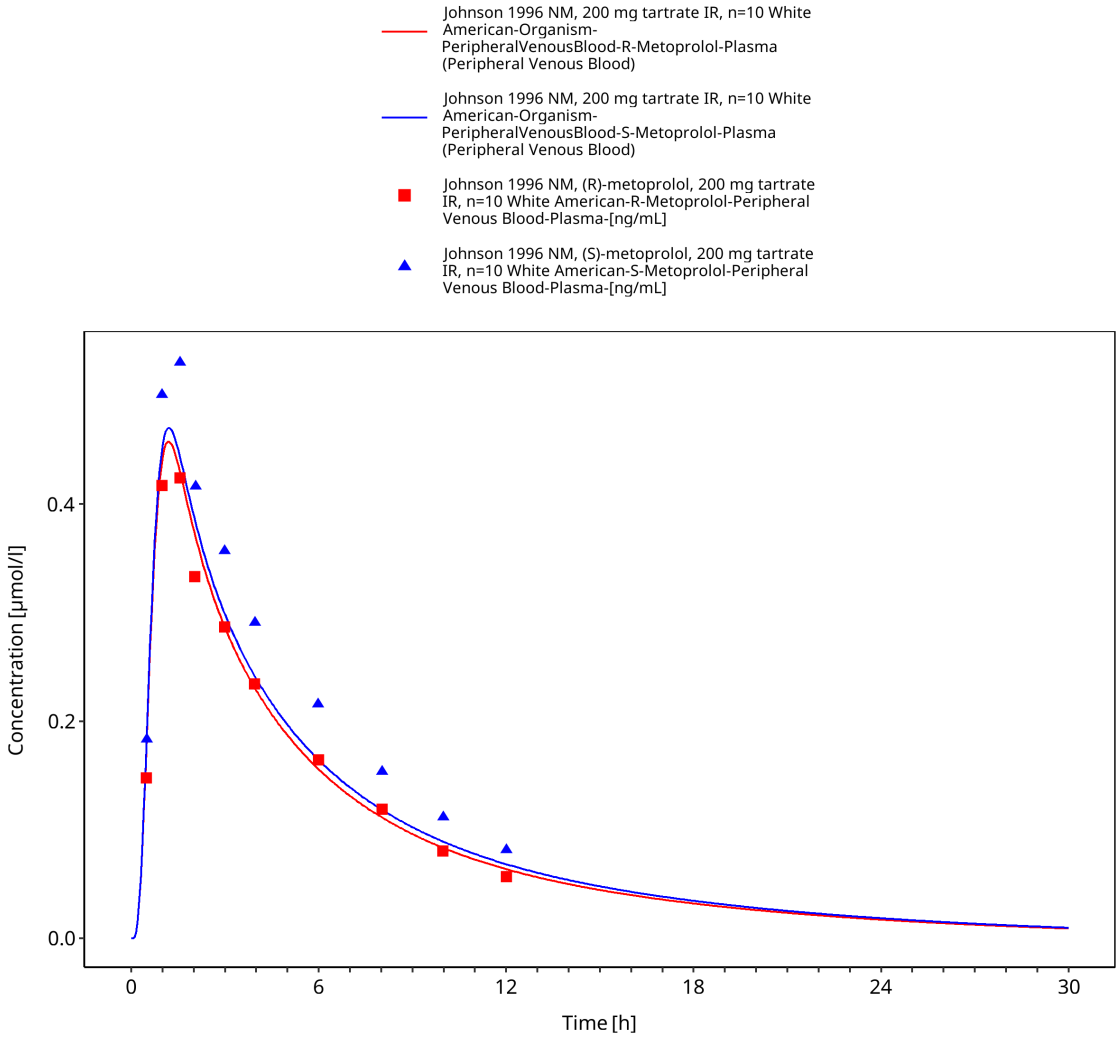


Figure 3-17: Johnson 1996: NM, 200 mg tartrate IR, n=10 White American

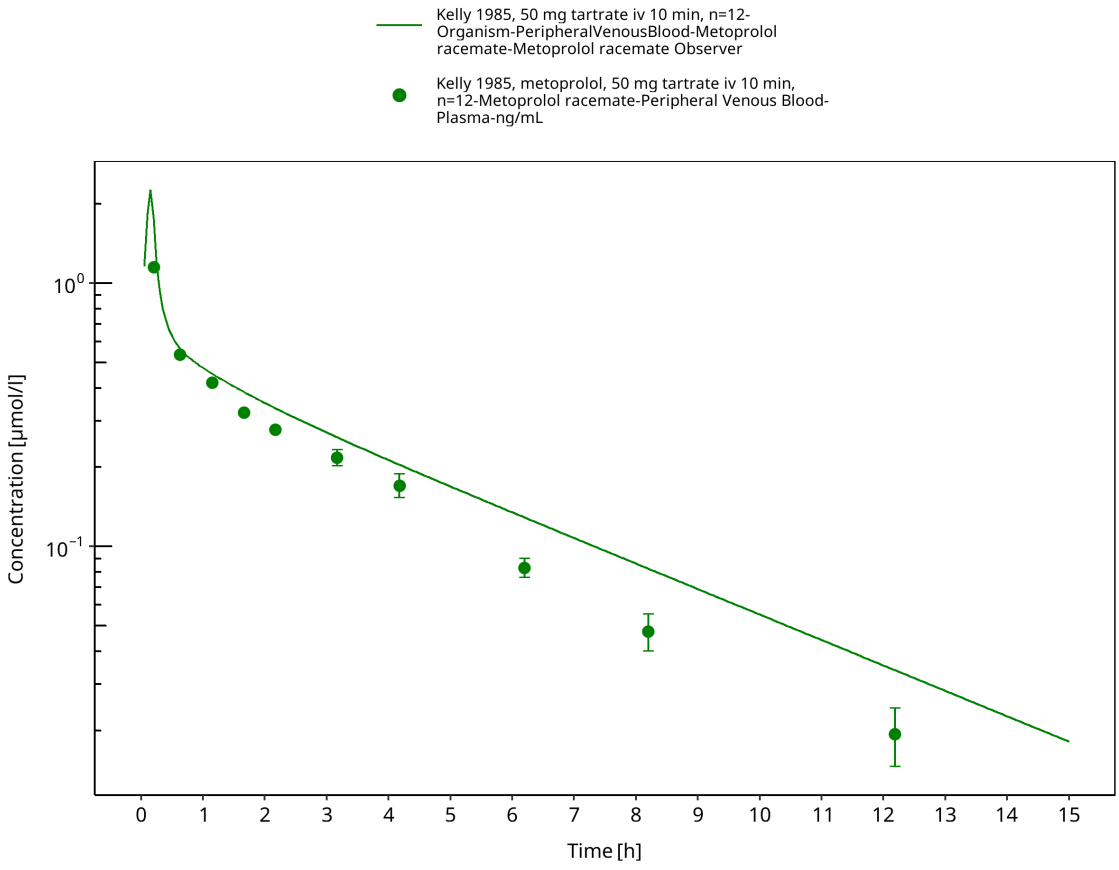


Figure 3-18: Kelly 1985: 50 mg tartrate iv 10 min, n=12

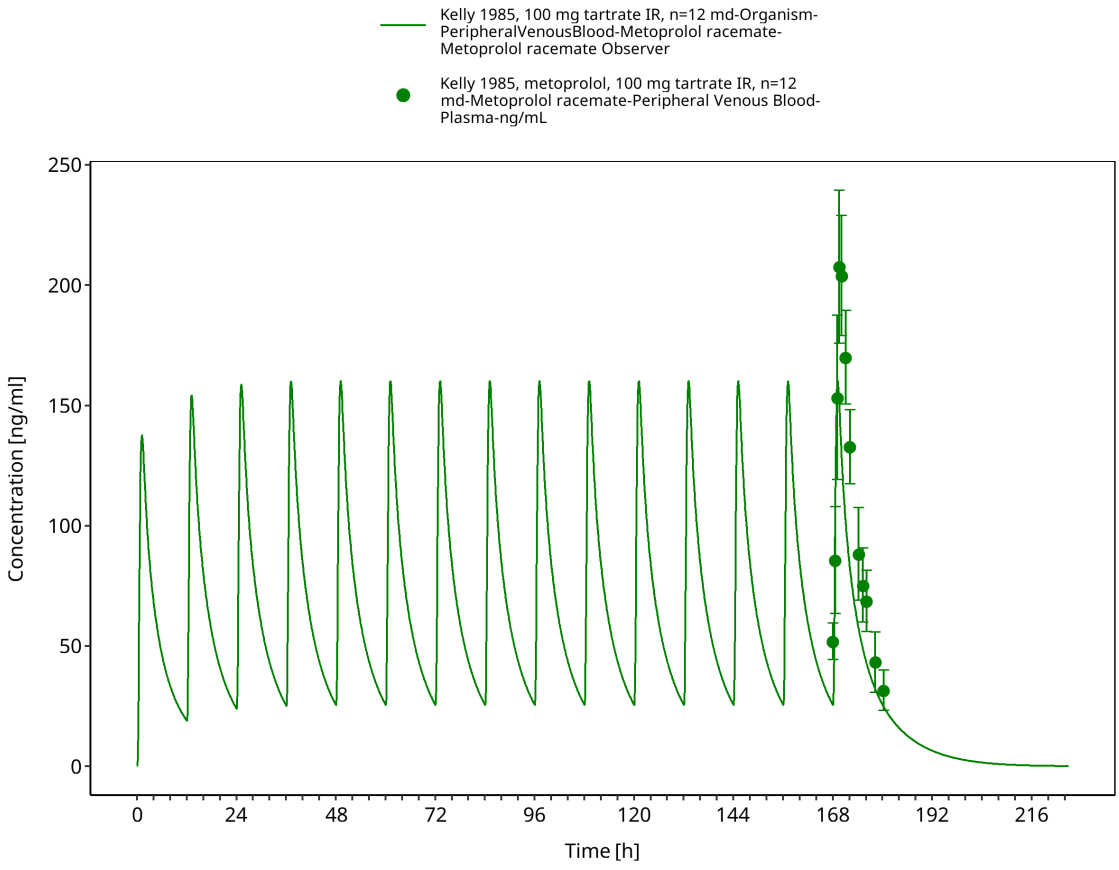


Figure 3-19: Kelly 1985: 100 mg b.i.d. tartrate IR, n=12, multiple dose

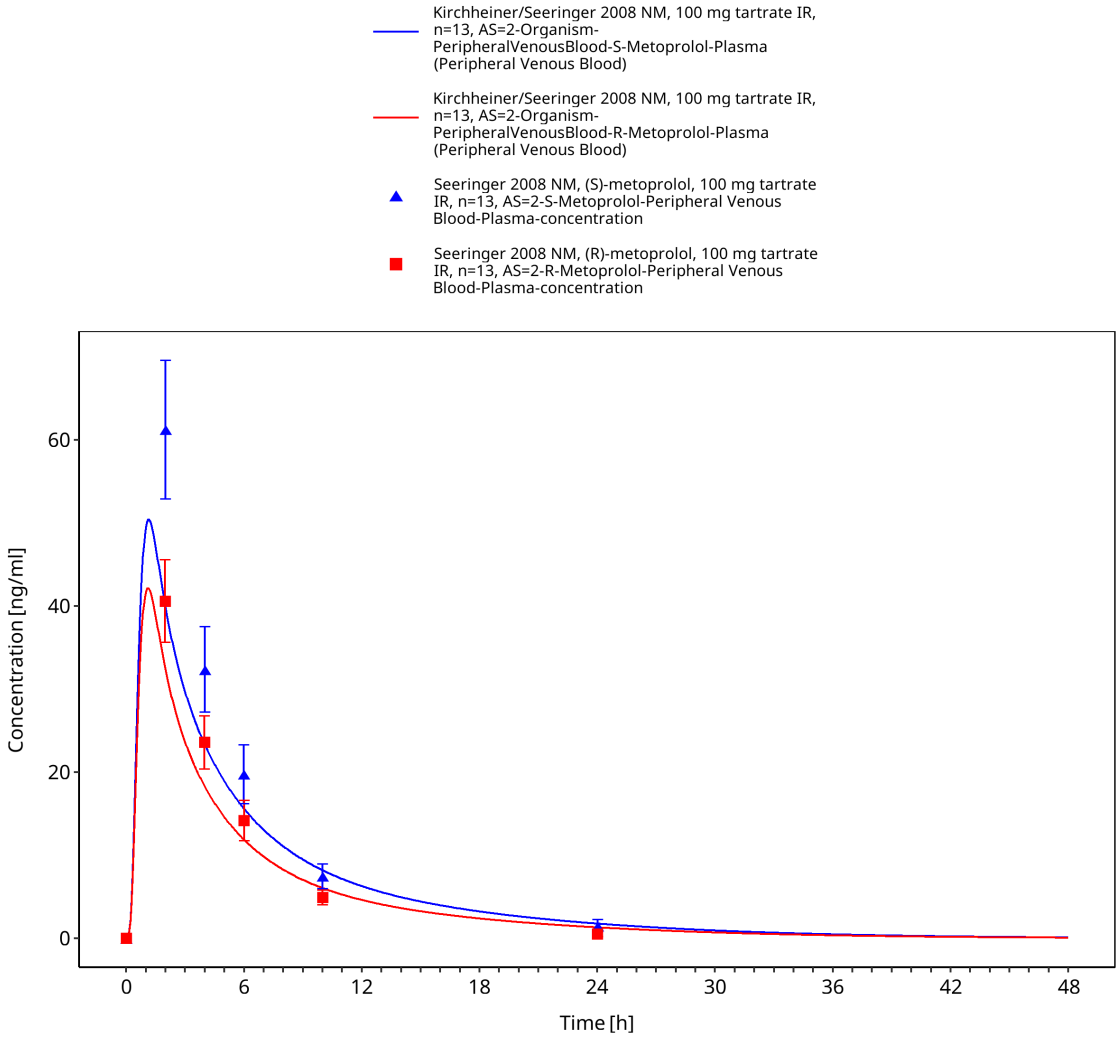


Figure 3-20: Kirchheiner 2004 / Seeringer 2008: NM, 100 mg tartrate IR, n=13, AS=2

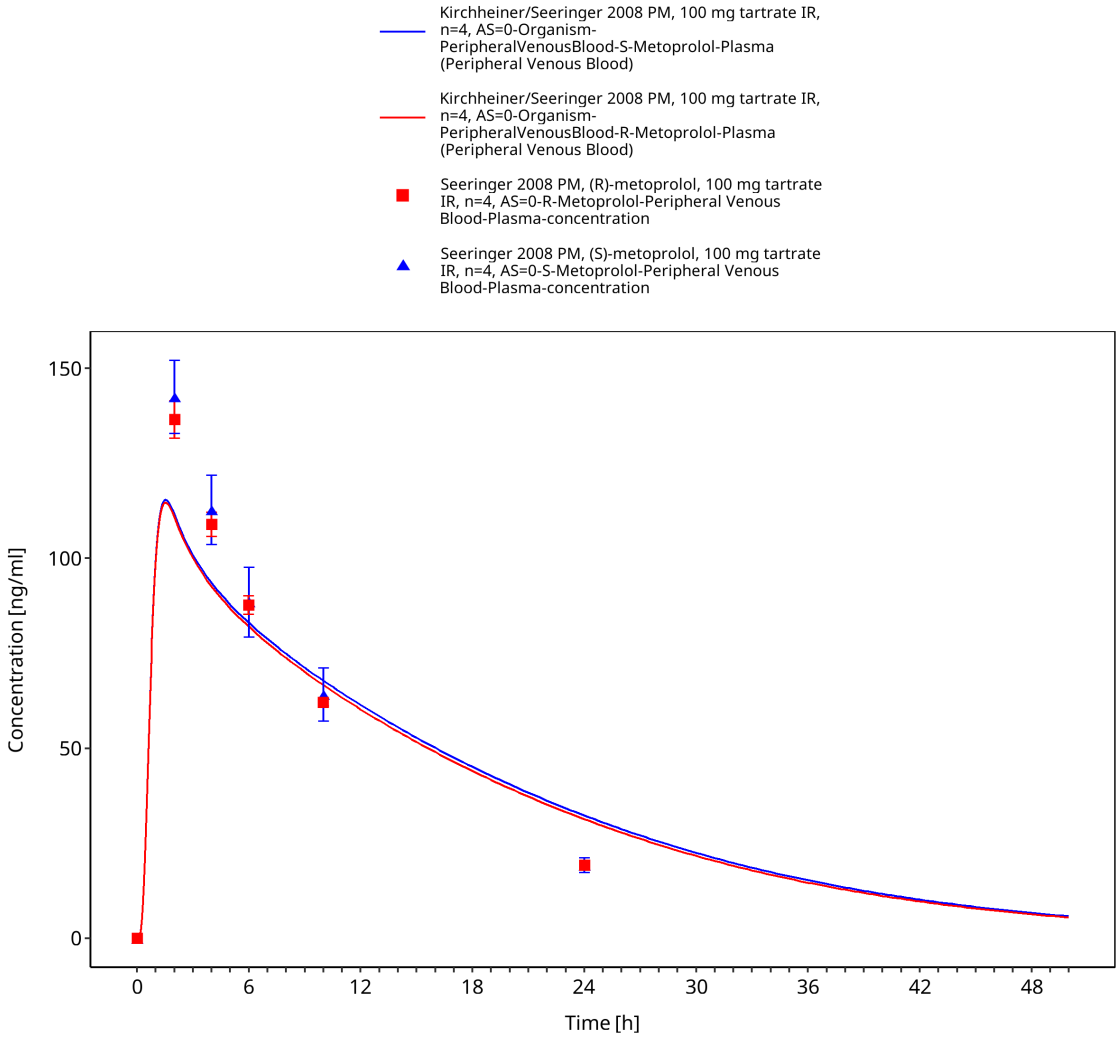


Figure 3-21: Kirchheiner 2004 / Seeringer 2008: PM, 100 mg tartrate IR, n=4, AS=0

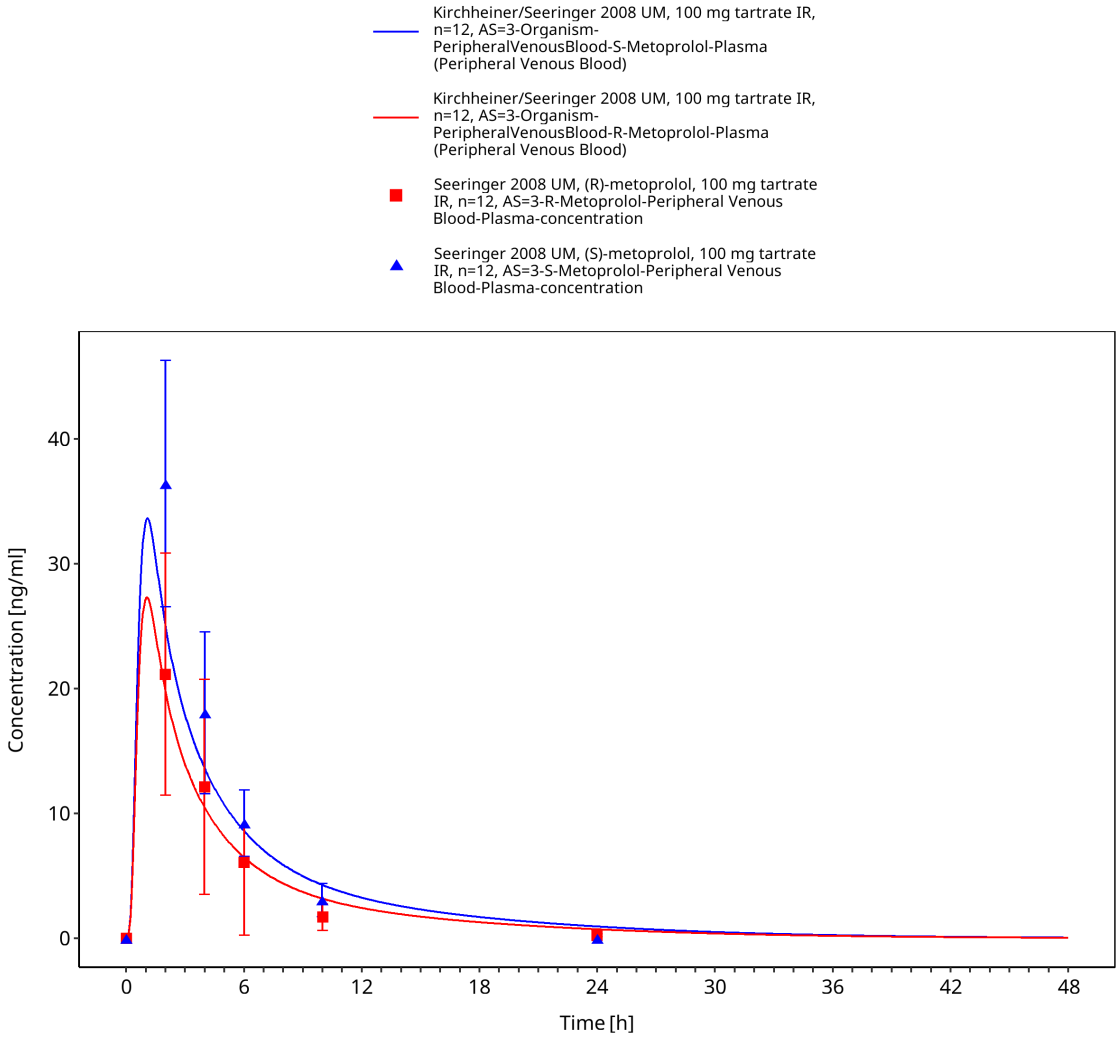


Figure 3-22: Kirchheiner 2004 / Seeringer 2008: UM, 100 mg tartrate IR, n=12, AS=3

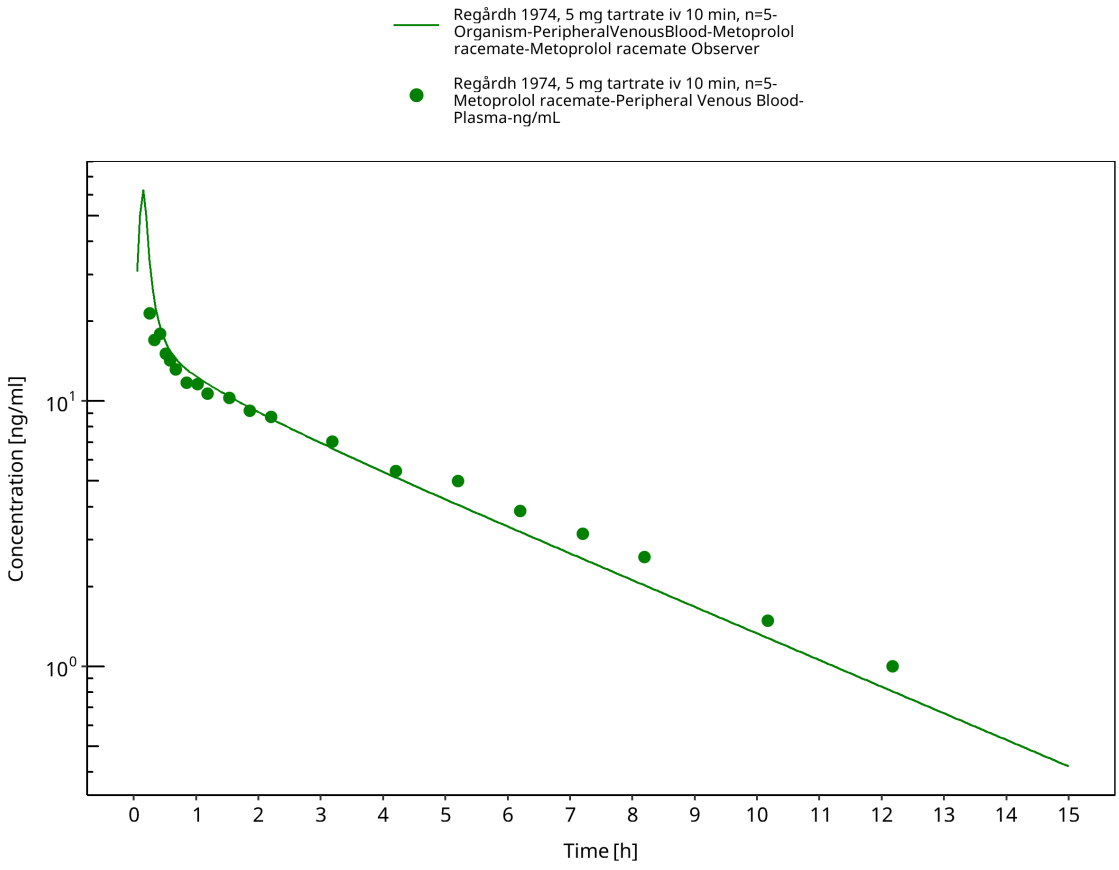


Figure 3-23: Regårdh 1974: 5 mg tartrate iv 10 min, n=5

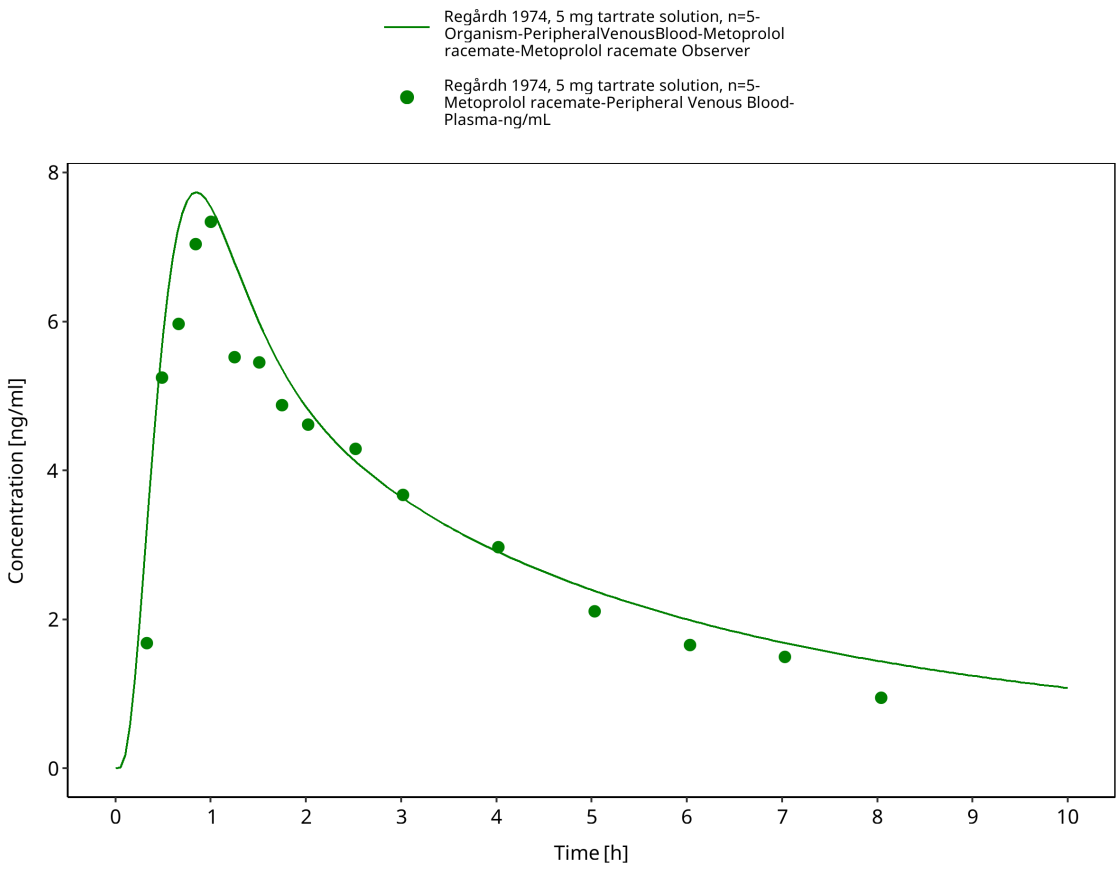


Figure 3-24: Regårdh 1974: 5 mg tartrate solution, n=5

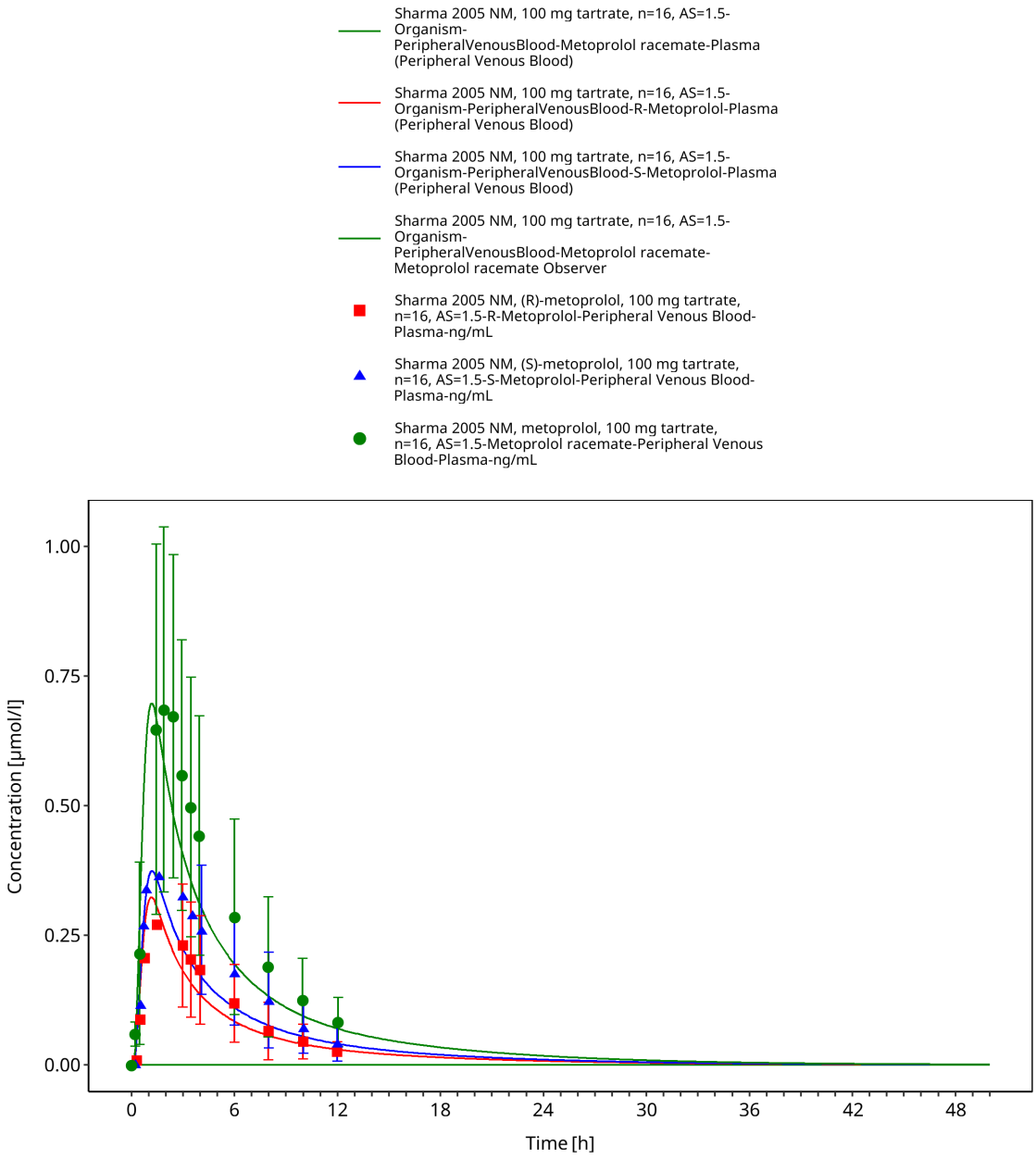


Figure 3-25: Sharma 2005: NM, 100 mg tartrate, n=16, AS=1.5

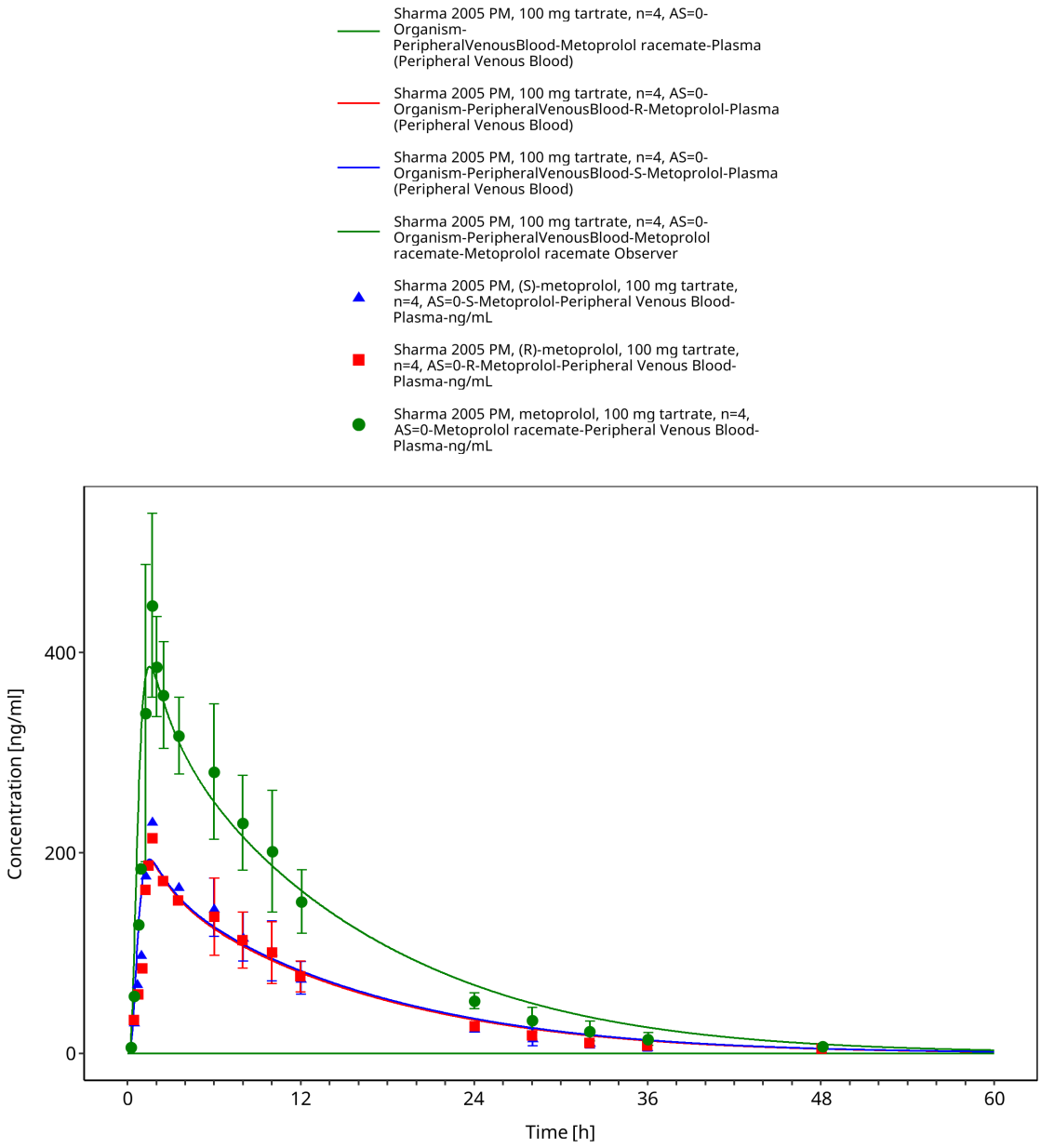


Figure 3-26: Sharma 2005: PM, 100 mg tartrate, n=4, AS=0

3.3.2 Model Verification

Model-verification profiles include additional adult clinical studies used to assess model behavior across sex, formulation and CYP2D6 status conditions not used as the primary fitting dataset. They also include seven metoprolol-only control arms of the CYP2D6 interaction network (Rüdesheim 2025).

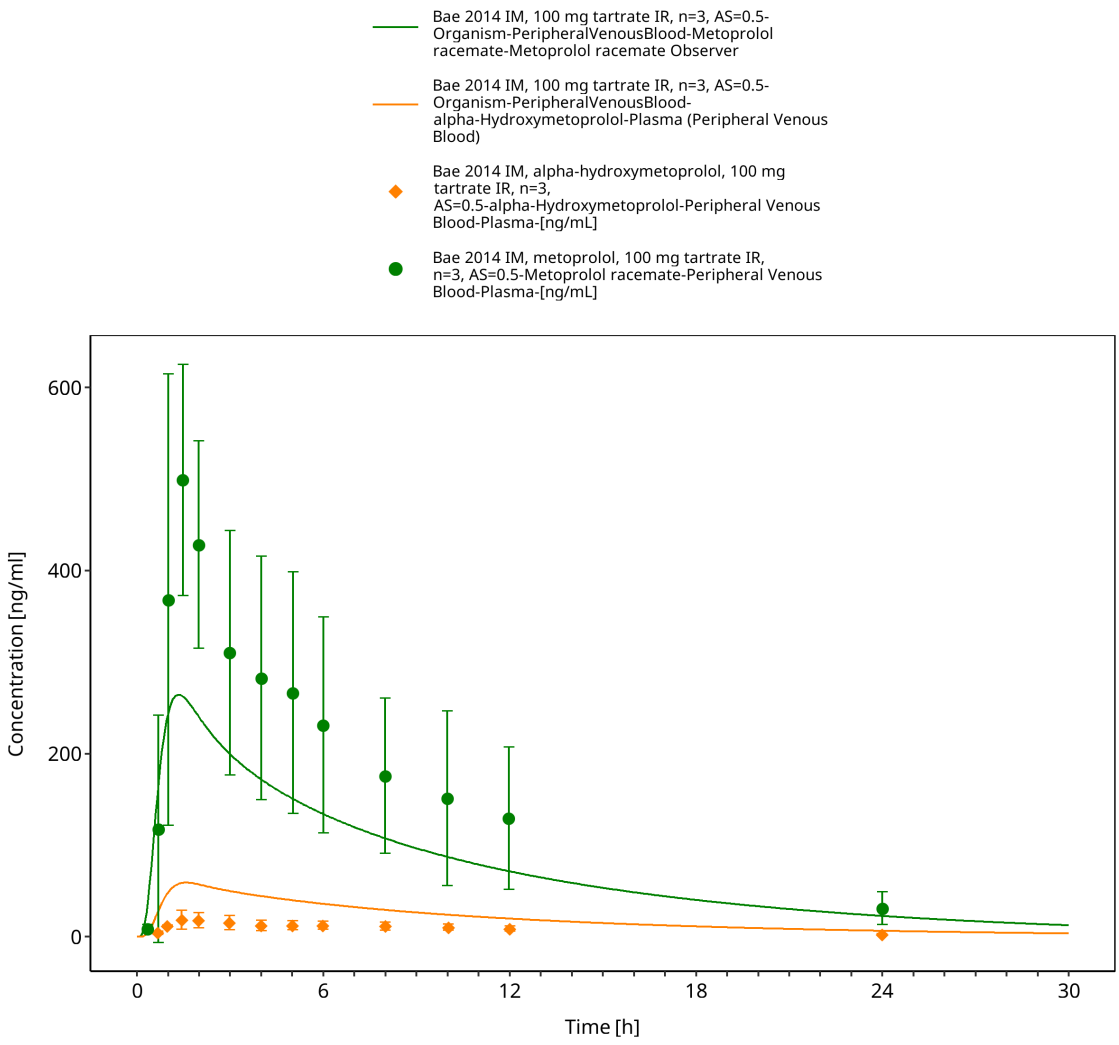


Figure 3-27: Bae 2014: IM, 100 mg tartrate IR, n=3, AS=0.5

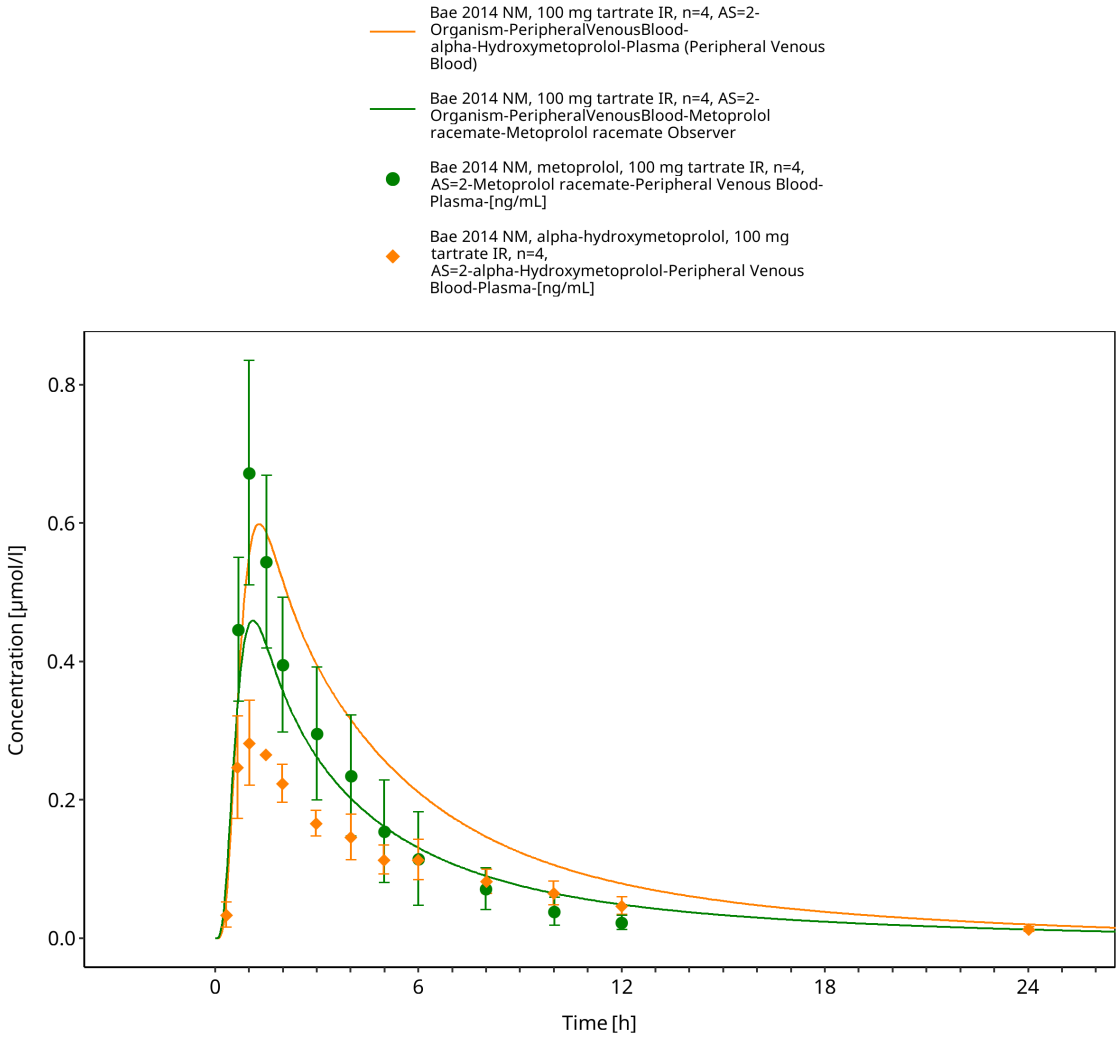


Figure 3-28: Bae 2014: NM, 100 mg tartrate IR, n=4, AS=2

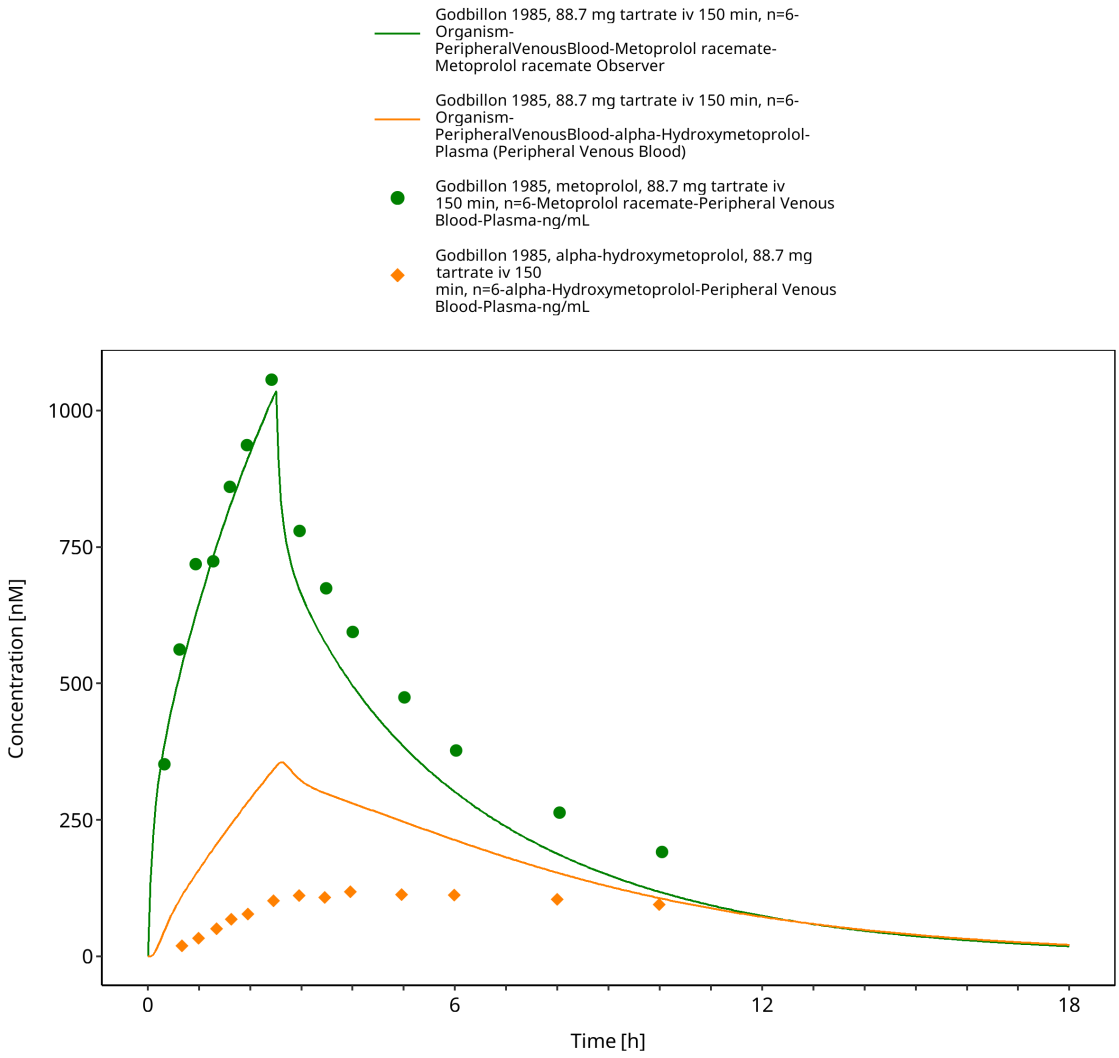


Figure 3-29: Godbillon 1985: 88.7 mg tartrate iv 150 min, n=6

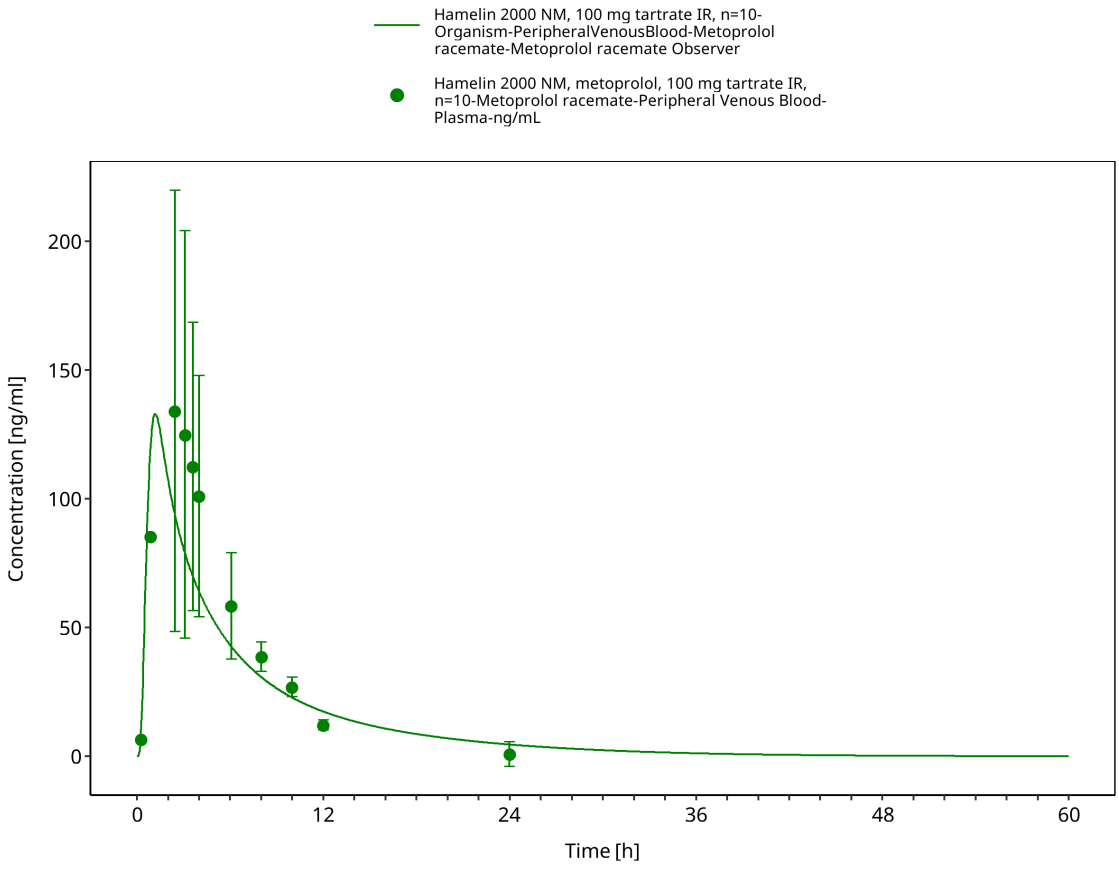


Figure 3-30: Hamelin 2000: NM, 100 mg tartrate IR, n=10

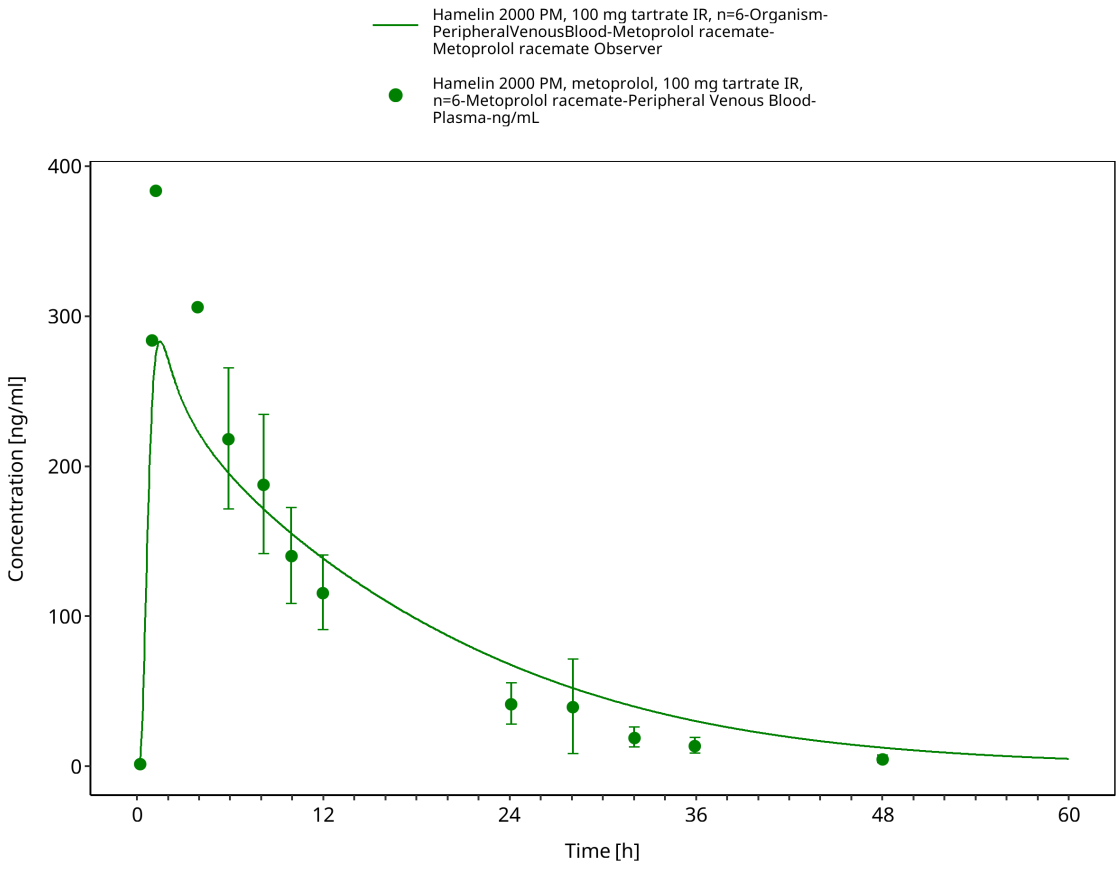


Figure 3-31: Hamelin 2000: PM, 100 mg tartrate IR, n=6

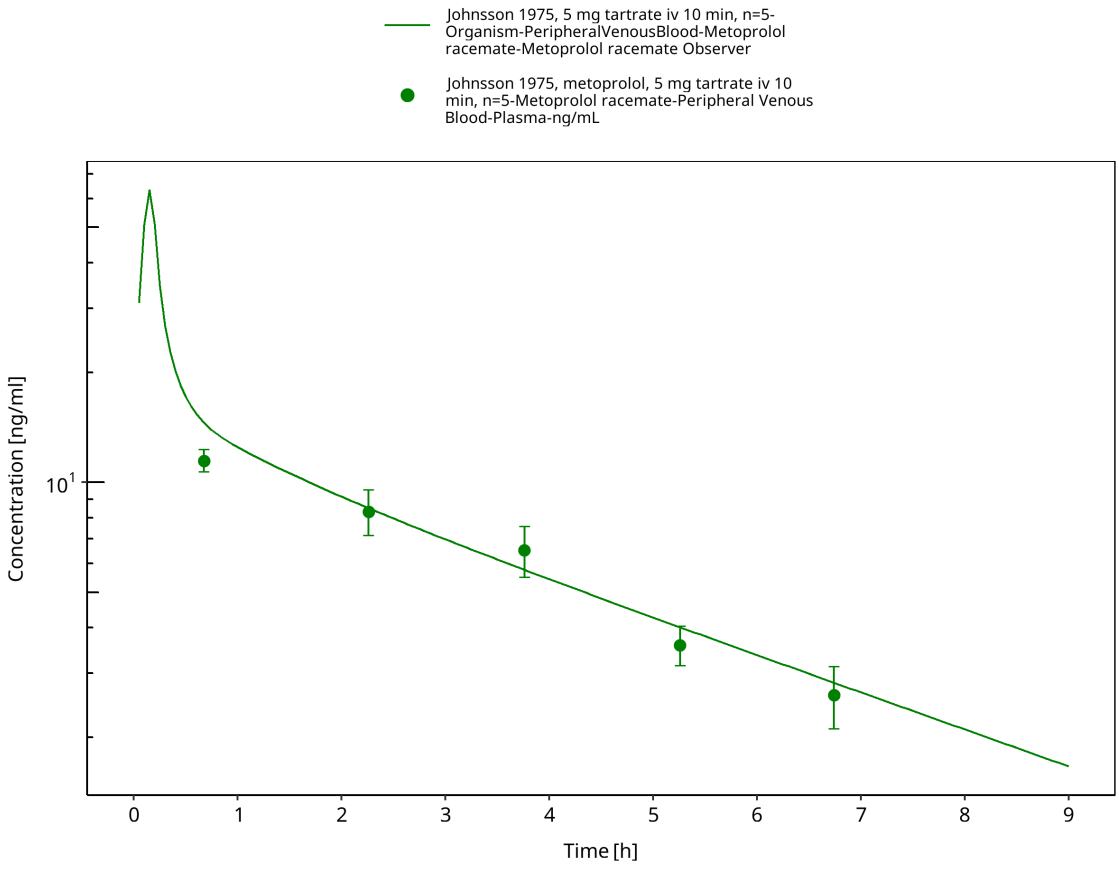


Figure 3-32: Johnsson 1975: 5 mg tartrate iv 10 min, n=5

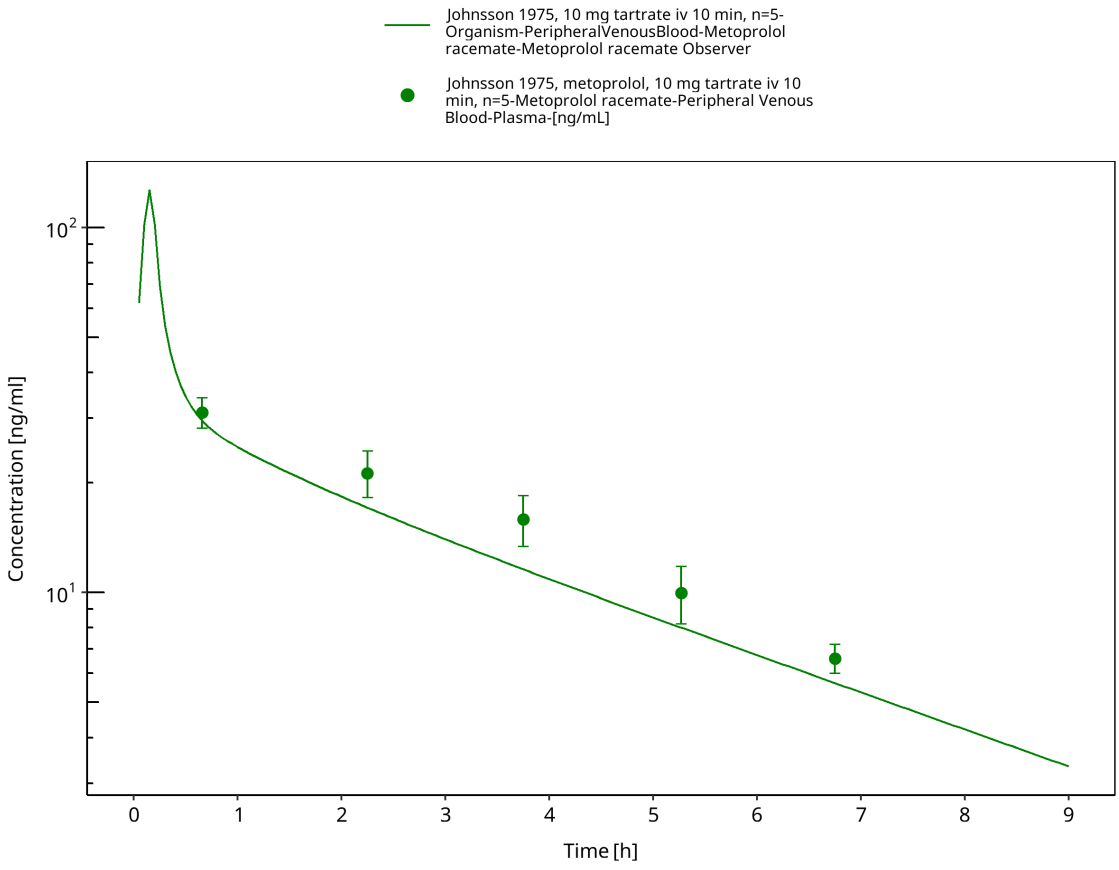


Figure 3-33: Johnsson 1975: 10 mg tartrate iv 10 min, n=5

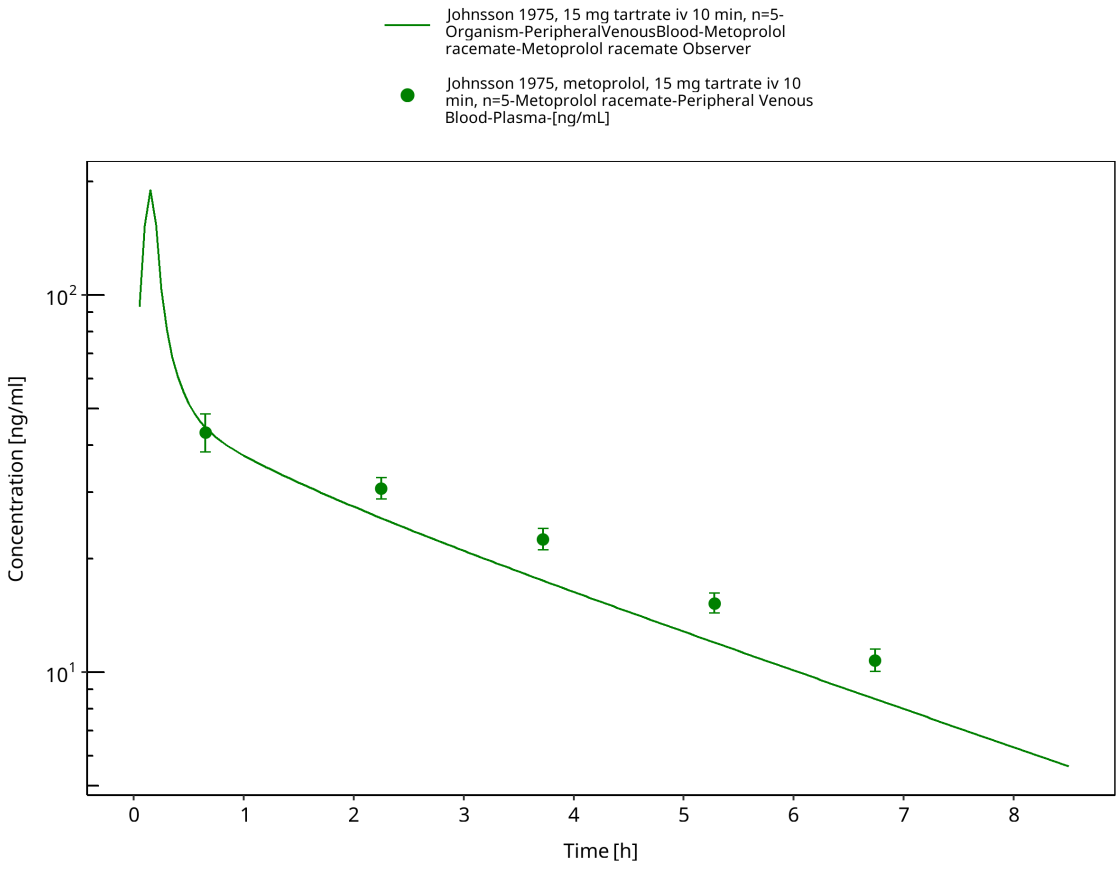


Figure 3-34: Johnsson 1975: 15 mg tartrate iv 10 min, n=5

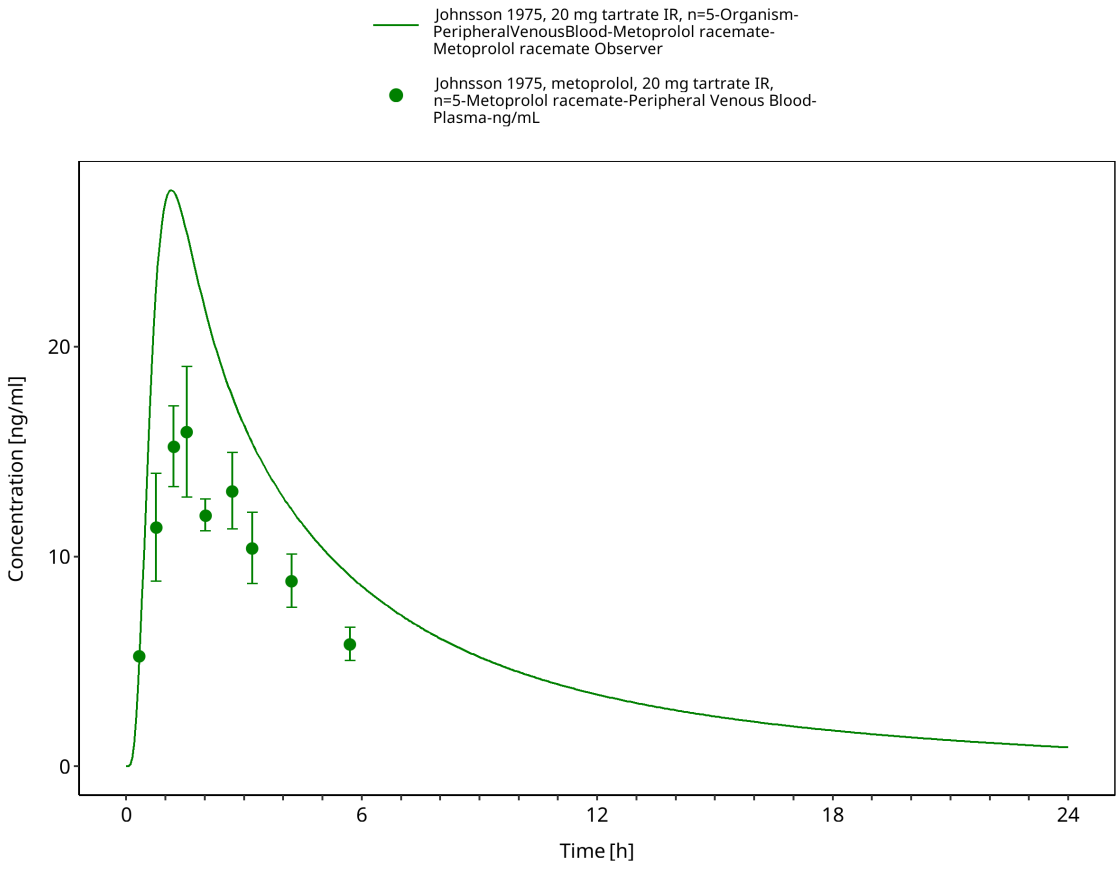


Figure 3-35: Johnsson 1975: 20 mg tartrate IR, n=5

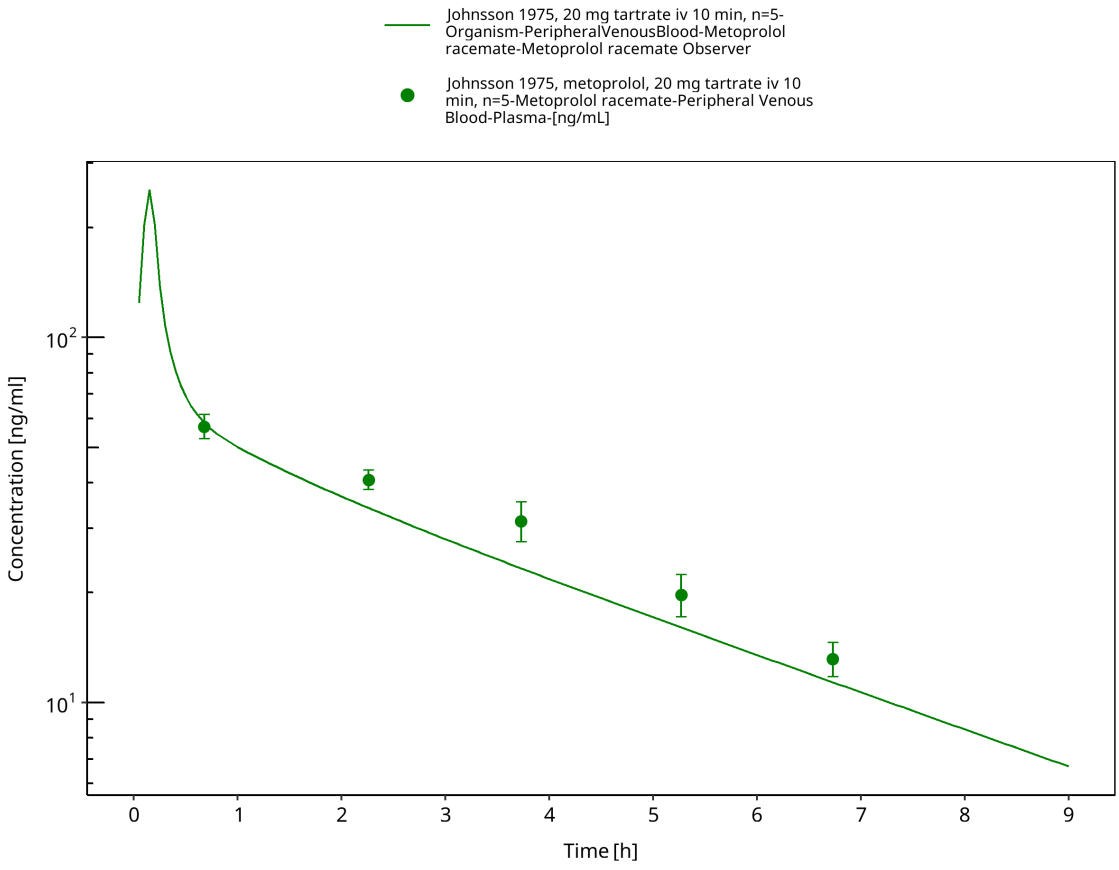


Figure 3-36: Johnsson 1975: 20 mg tartrate iv 10 min, n=5

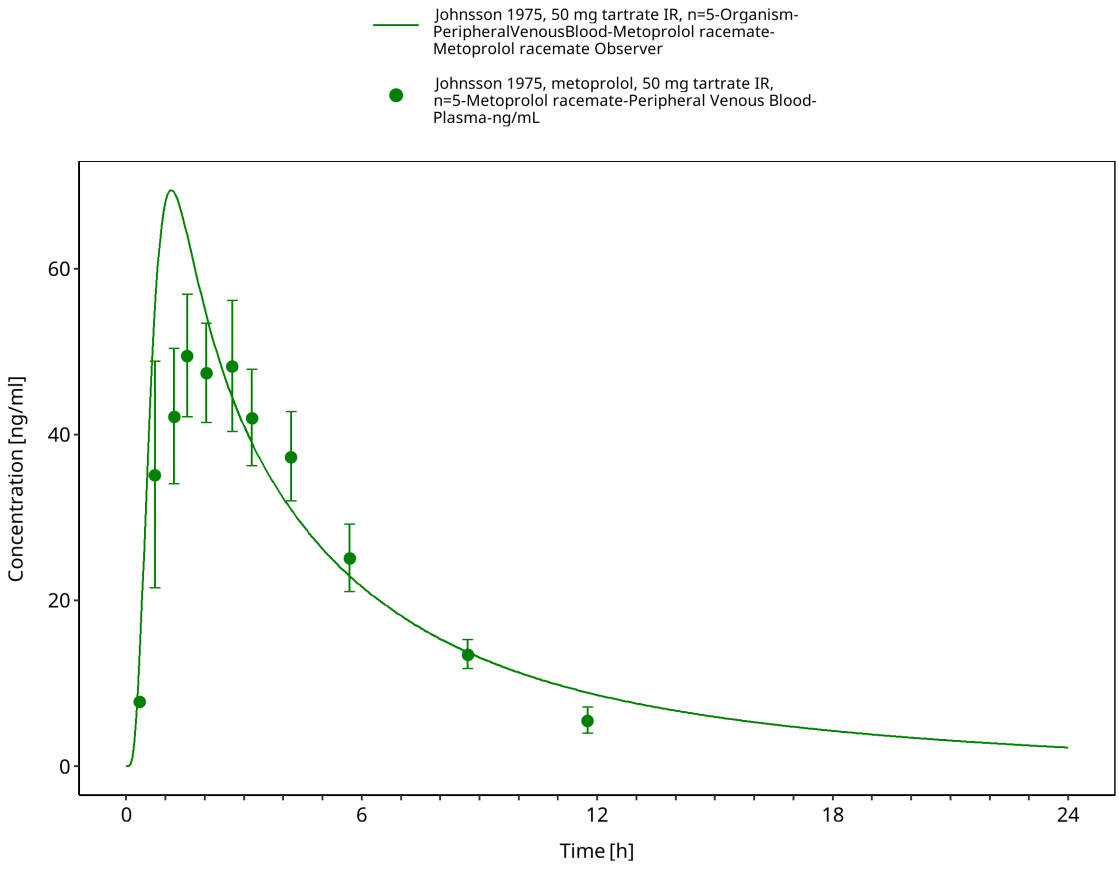


Figure 3-37: Johnsson 1975: 50 mg tartrate IR, n=5

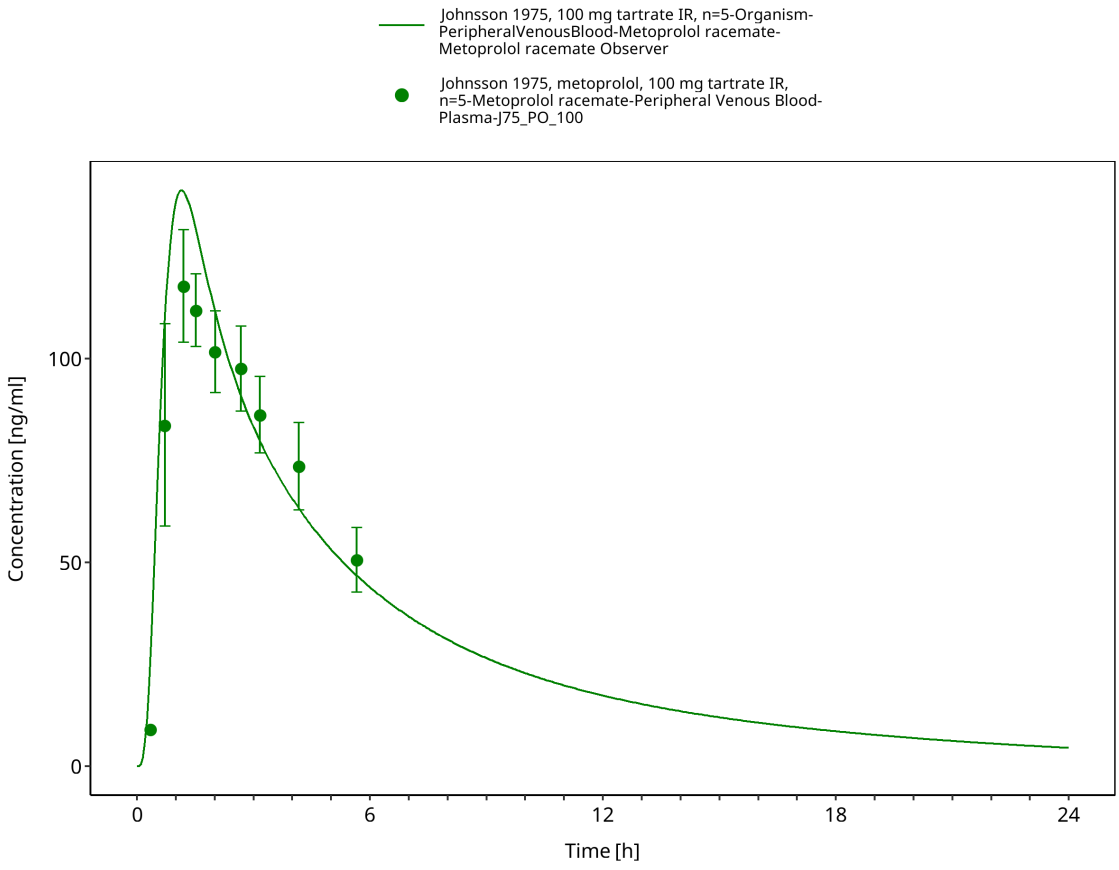


Figure 3-38: Johnsson 1975: 100 mg tartrate IR, n=5

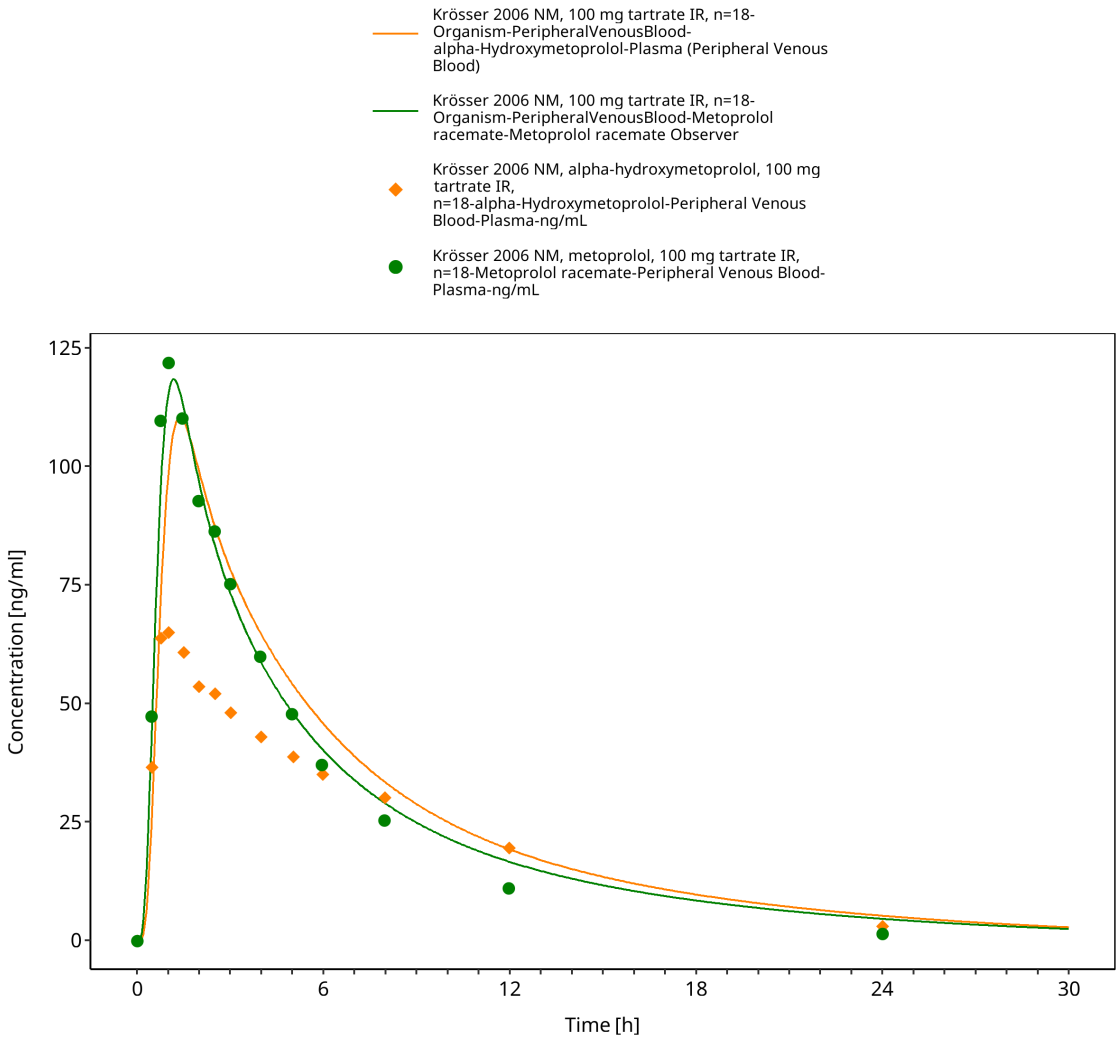


Figure 3-39: Krösser 2006: NM, 100 mg tartrate IR, n=18

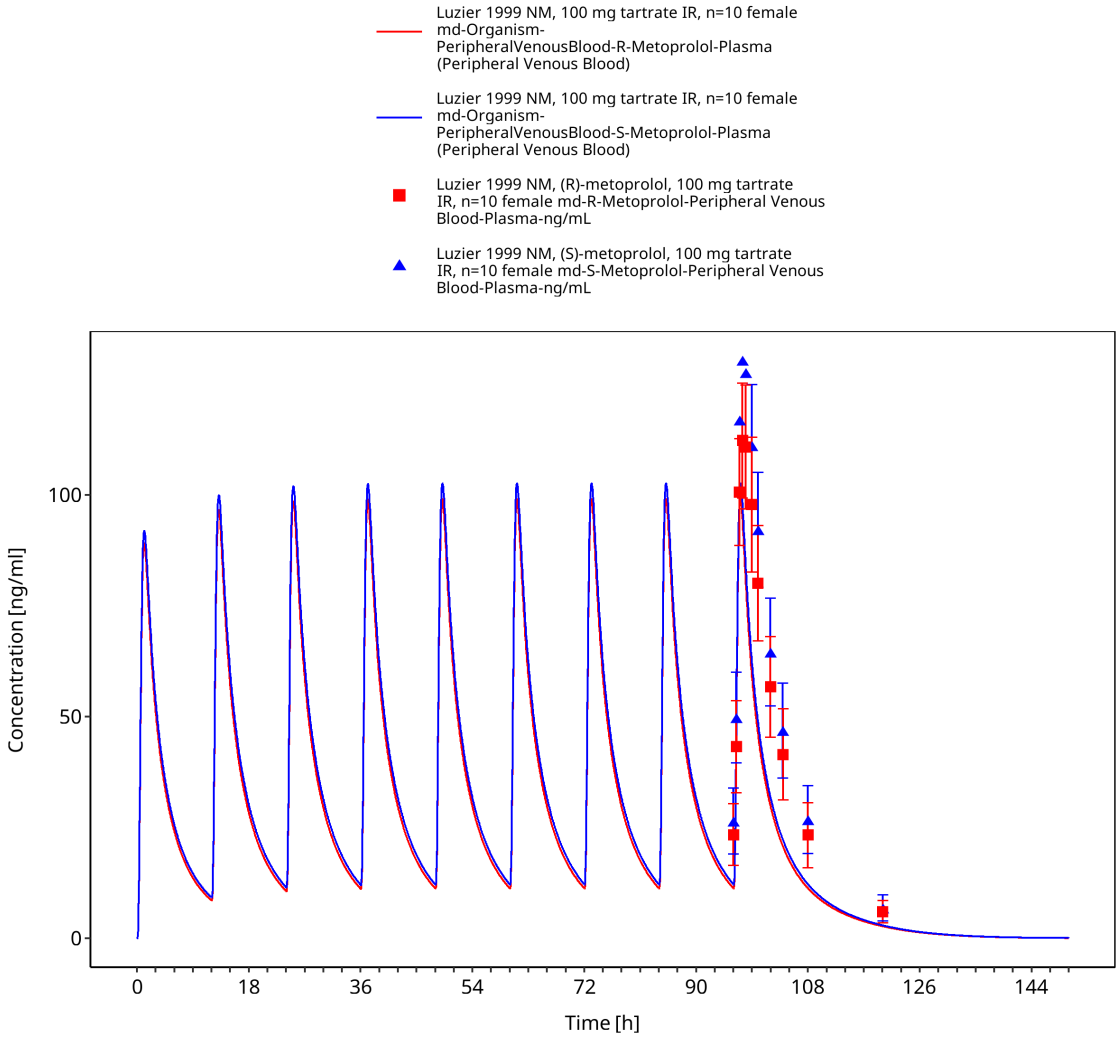


Figure 3-40: Luzier 1999: NM, 100 mg b.i.d. tartrate IR, n=10 female, multiple dose

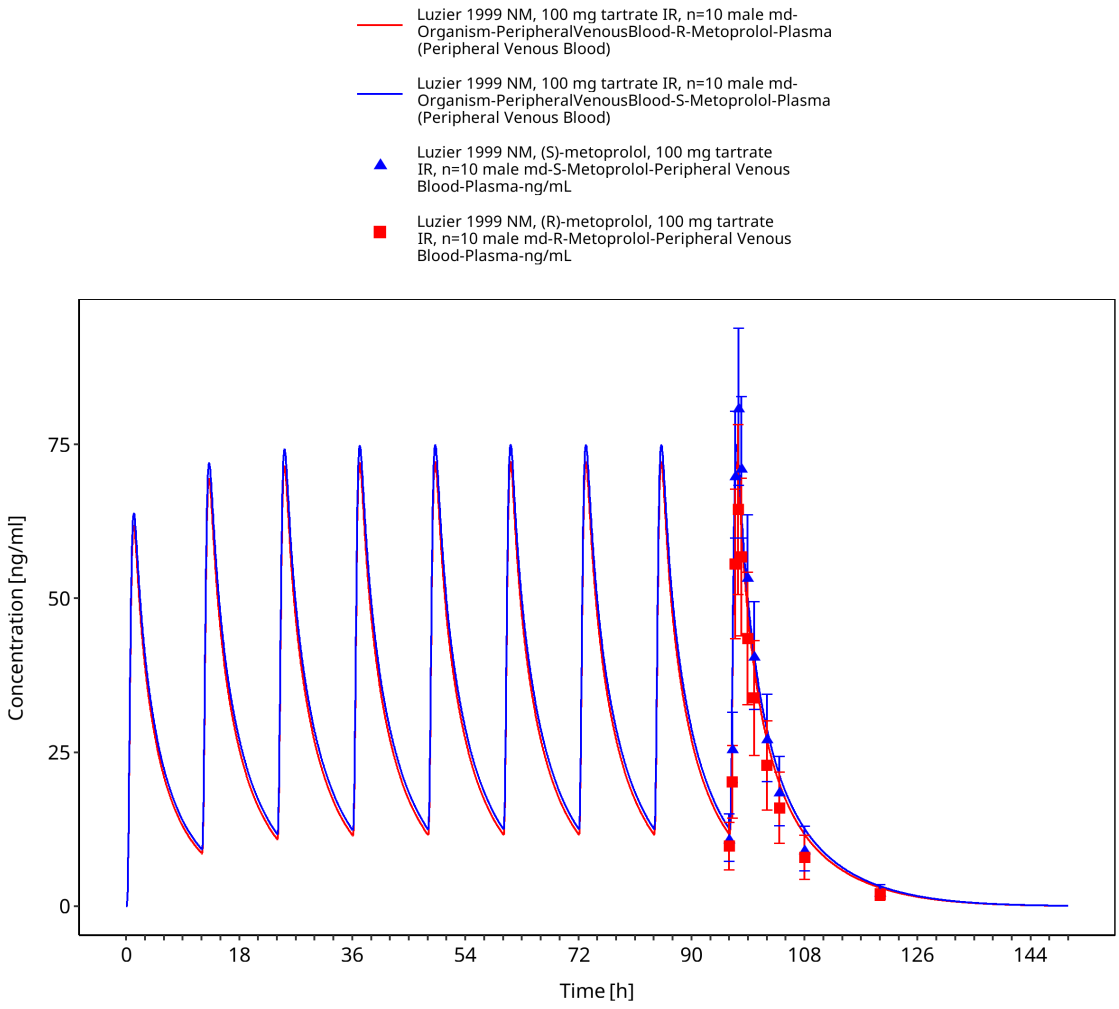


Figure 3-41: Luzier 1999: NM, 100 mg b.i.d. tartrate IR, n=10 male, multiple dose

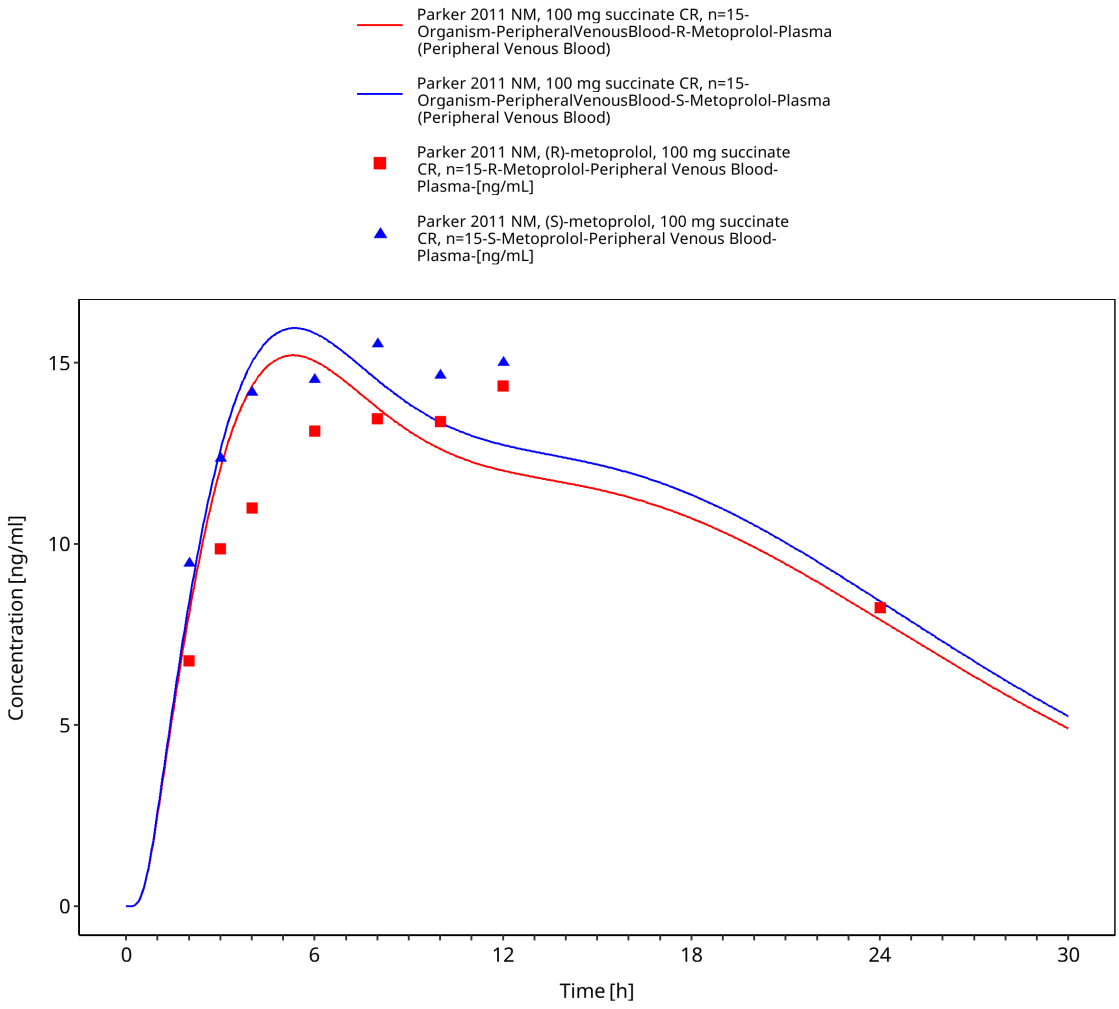


Figure 3-42: Parker 2011: NM, 100 mg succinate CR, n=15

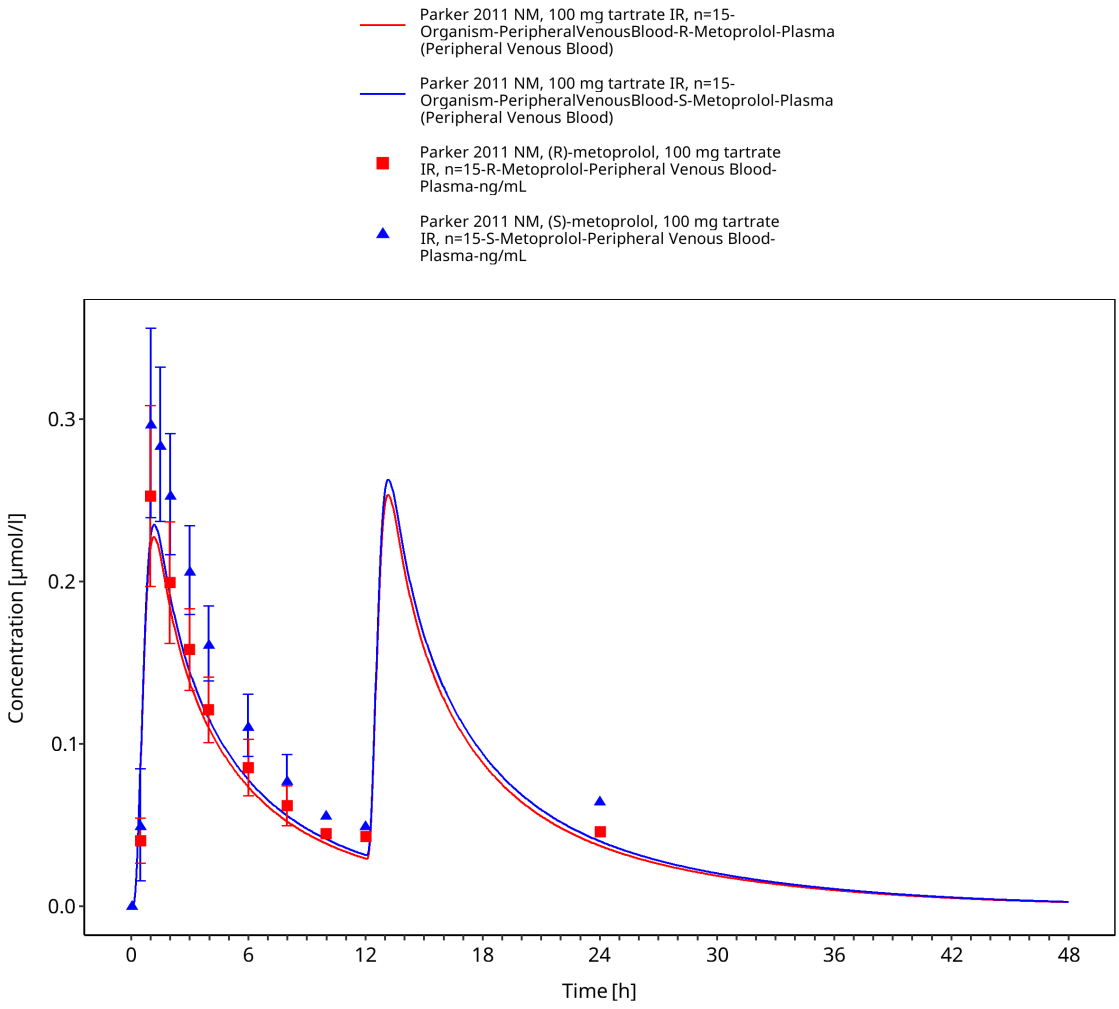


Figure 3-43: Parker 2011: NM, 100 mg tartrate IR, n=15

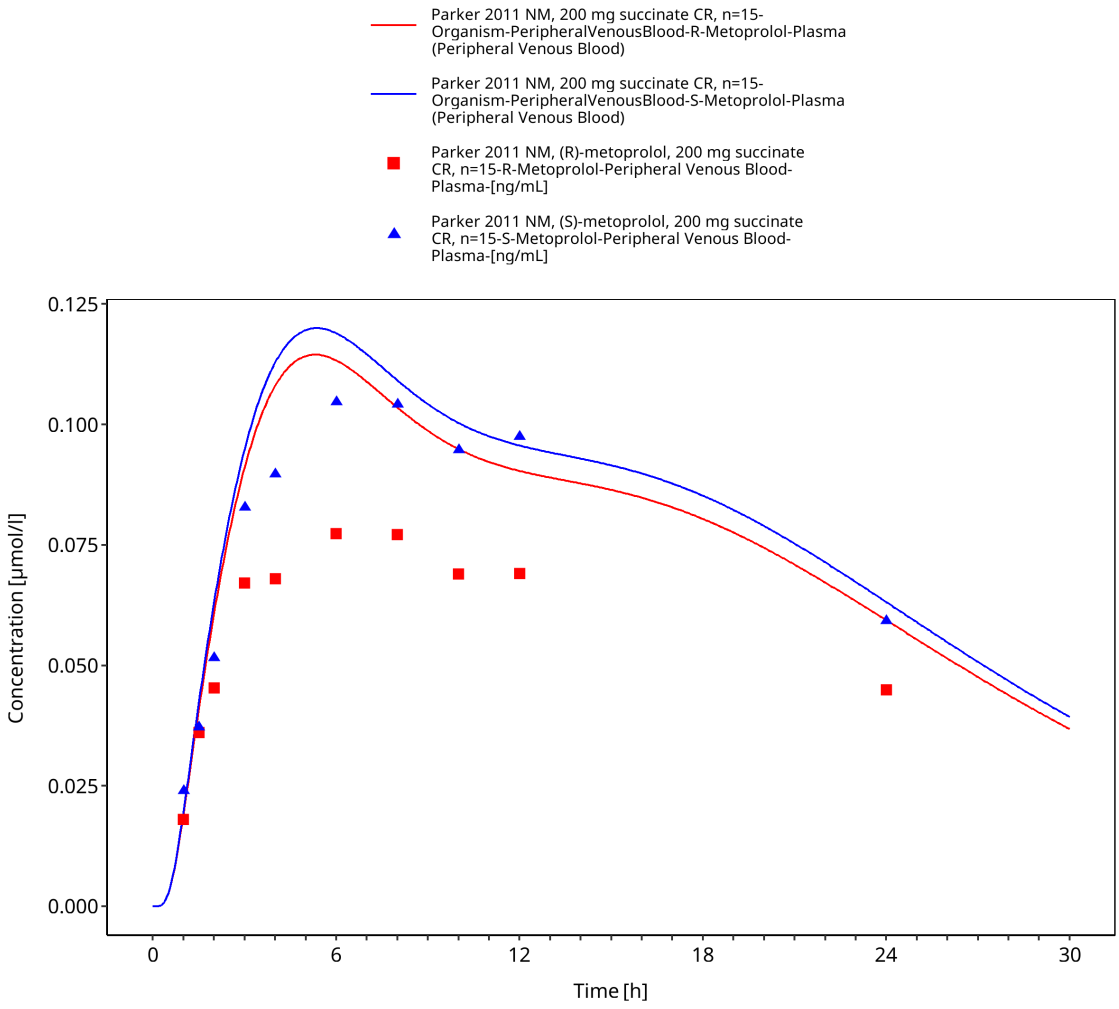


Figure 3-44: Parker 2011: NM, 200 mg succinate CR, n=15

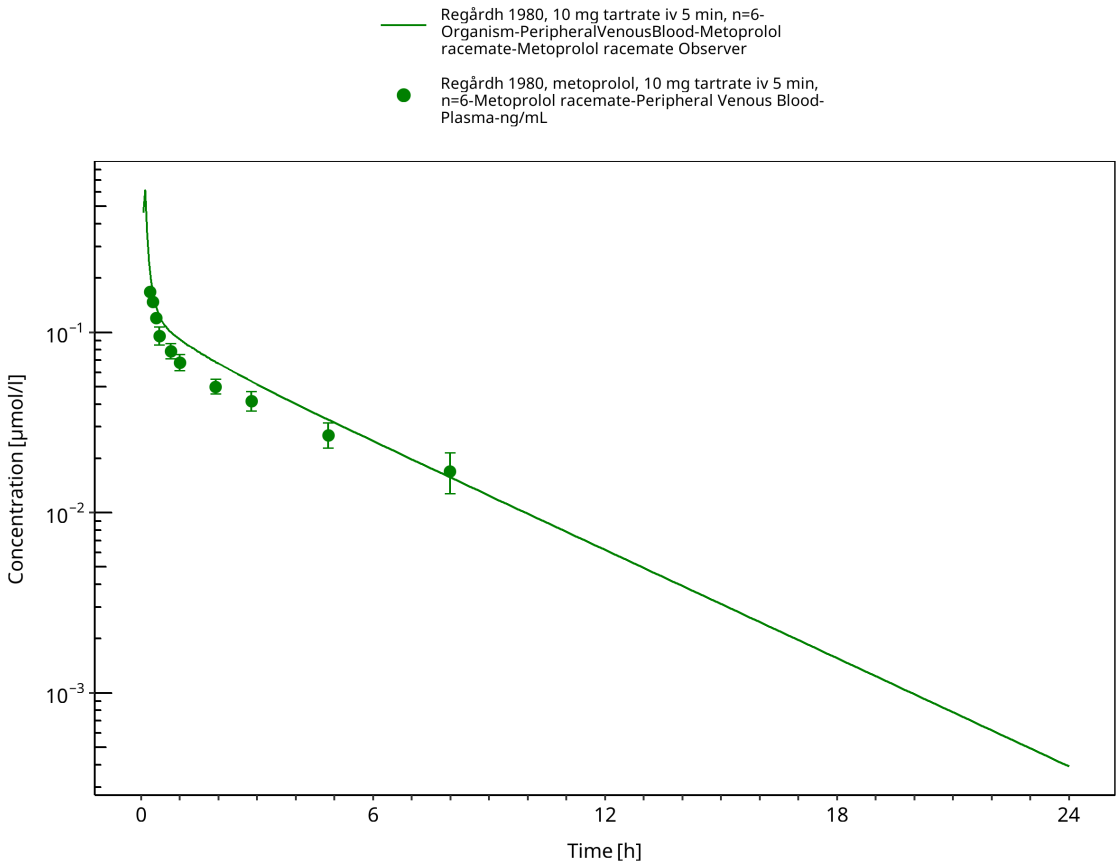


Figure 3-45: Regårdh 1980: 10 mg tartrate iv 5 min, n=6

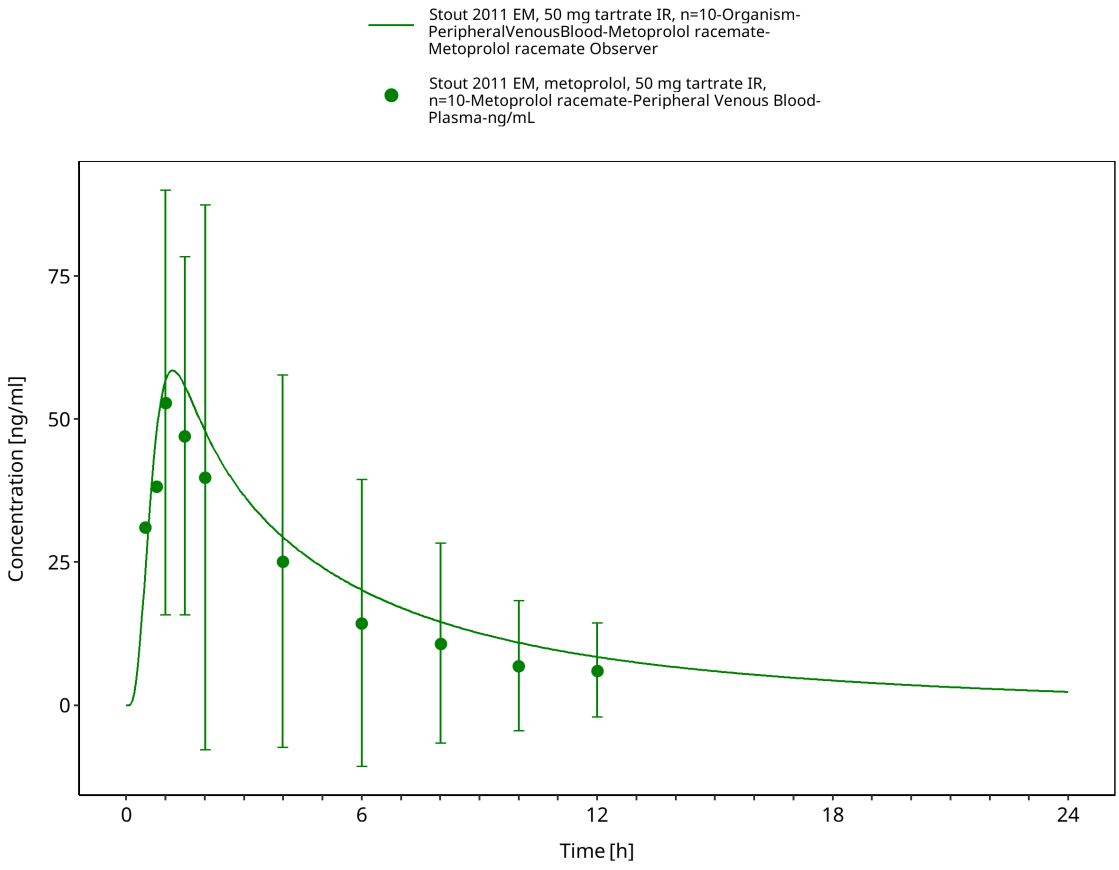


Figure 3-46: Stout 2011: EM, 50 mg tartrate IR, n=10

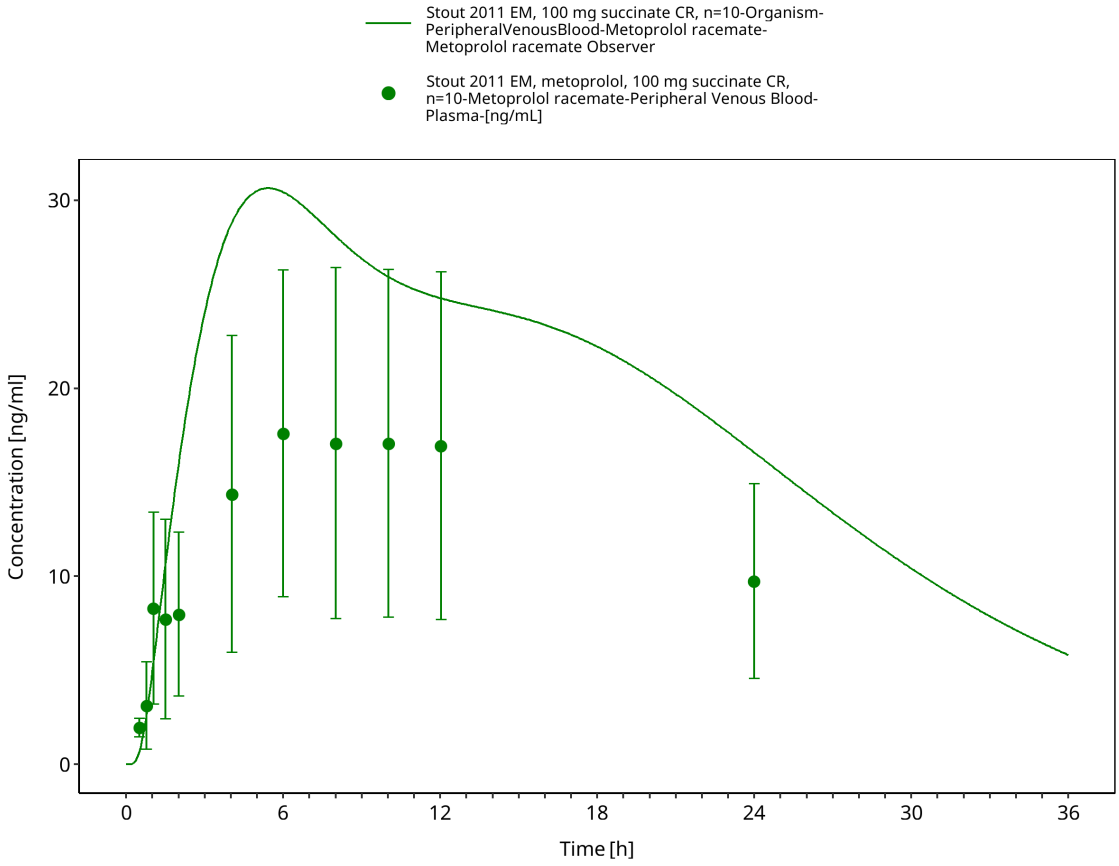


Figure 3-47: Stout 2011: EM, 100 mg succinate CR, n=10

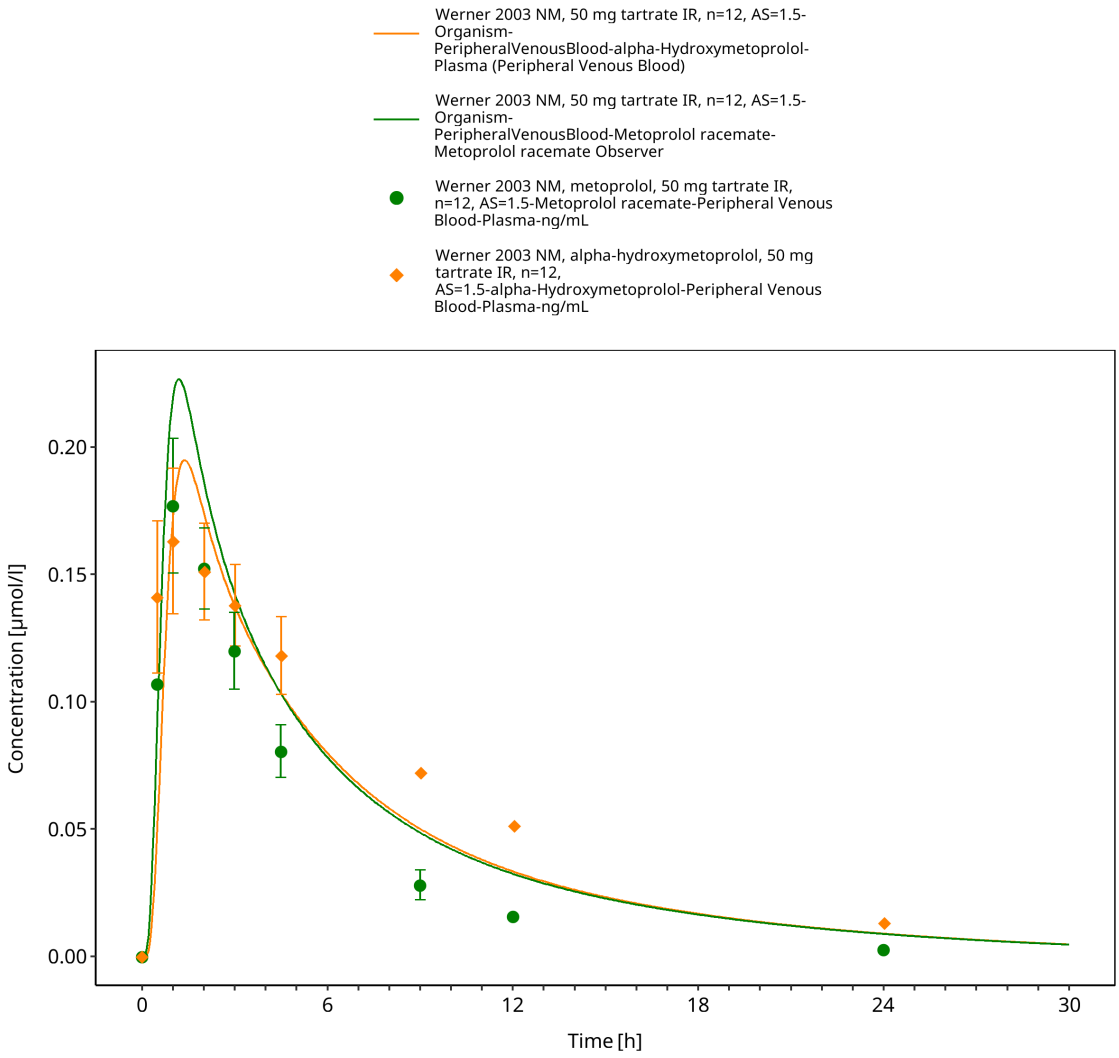


Figure 3-48: Werner 2003: NM, 50 mg tartrate IR, n=12, AS=1.5

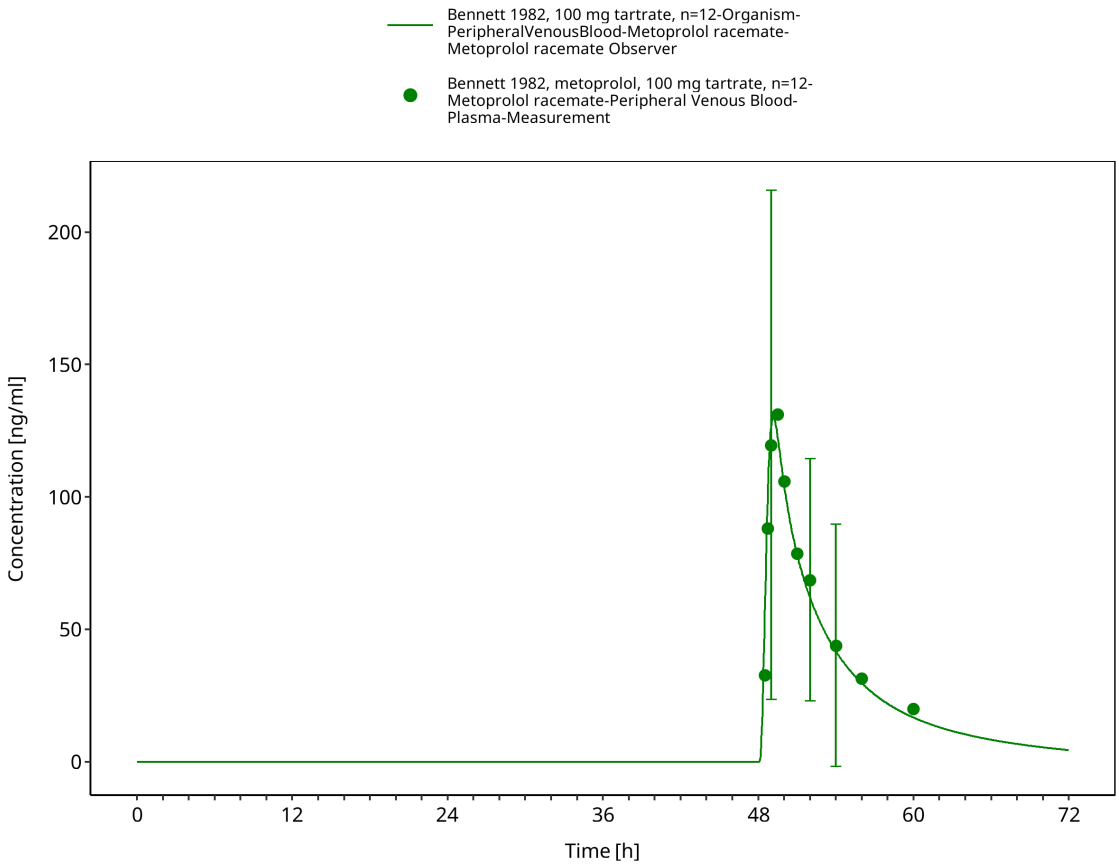


Figure 3-49: Bennett 1982: 100 mg tartrate, n=12

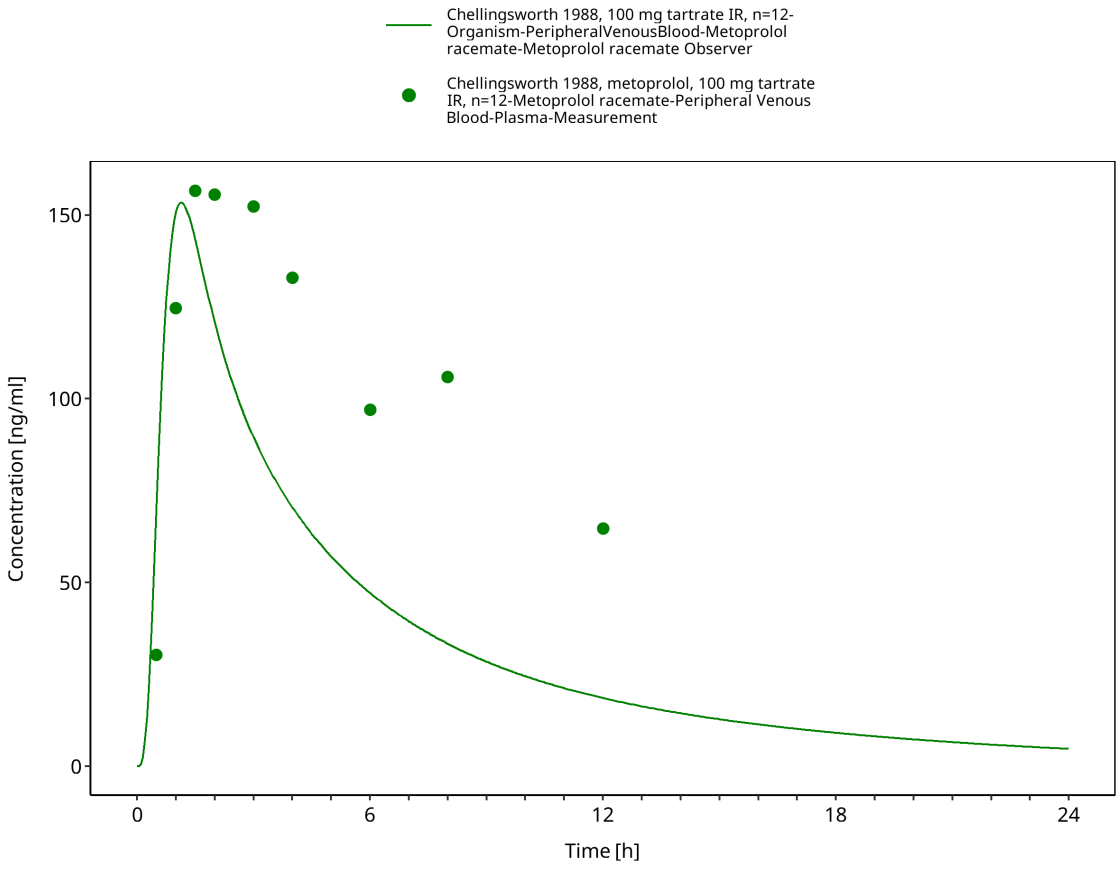


Figure 3-50: Chellingsworth 1988: 100 mg tartrate IR, n=12

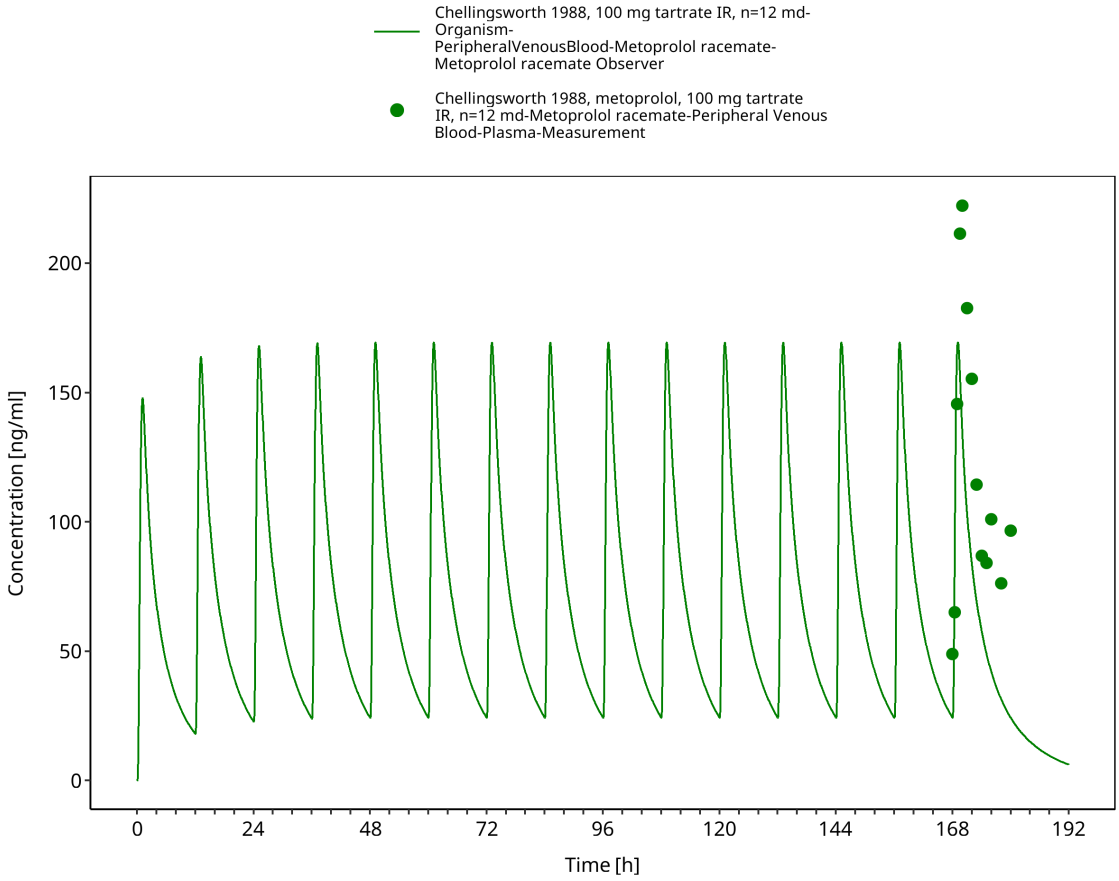


Figure 3-51: Chellingsworth 1988: 100 mg b.i.d. tartrate IR, n=12, multiple dose

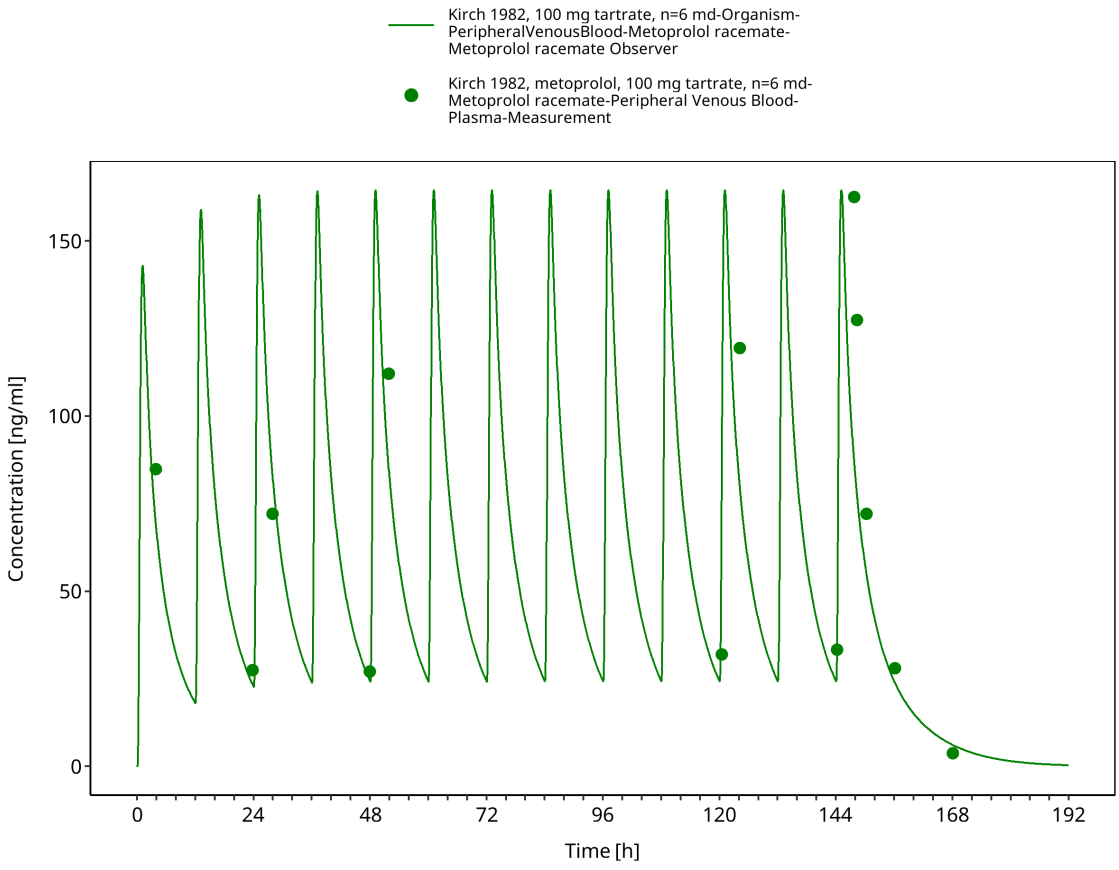


Figure 3-52: Kirch 1982: 100 mg b.i.d. tartrate, n=6, multiple dose

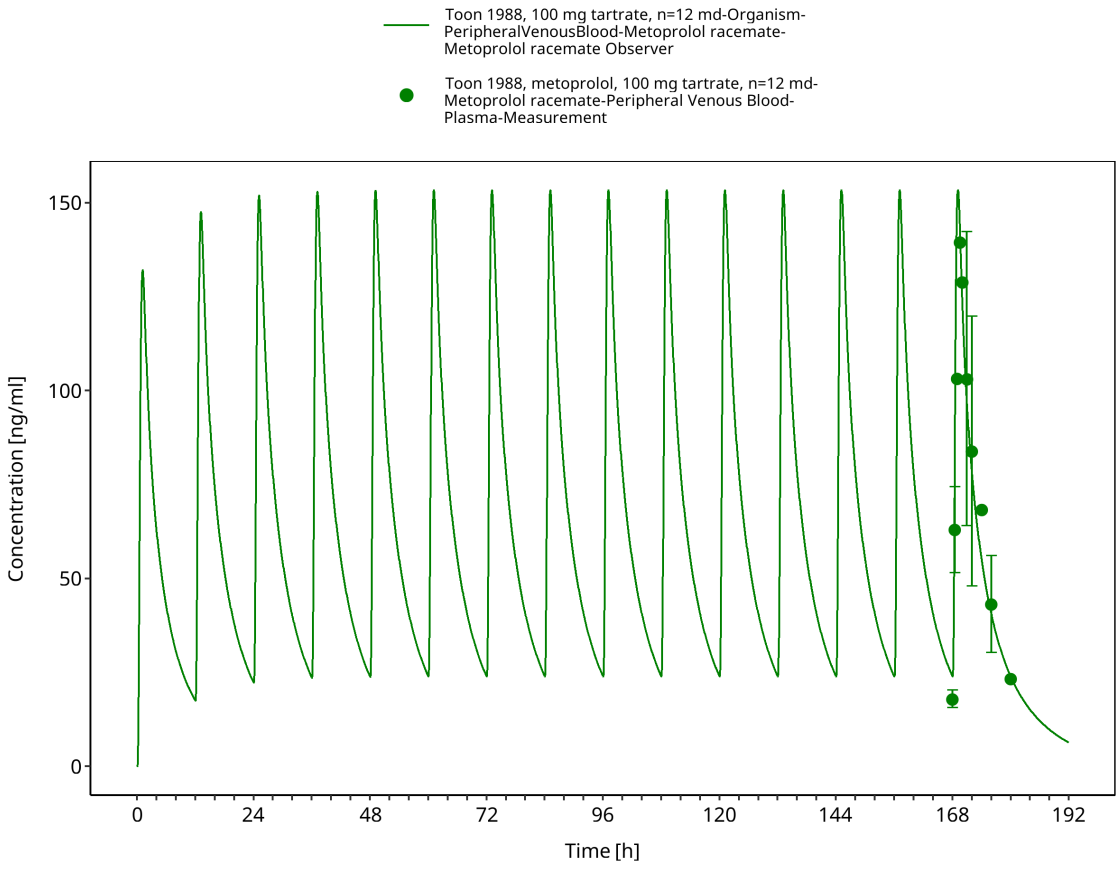


Figure 3-53: Toon 1988: 100 mg b.i.d. tartrate, n=12, multiple dose

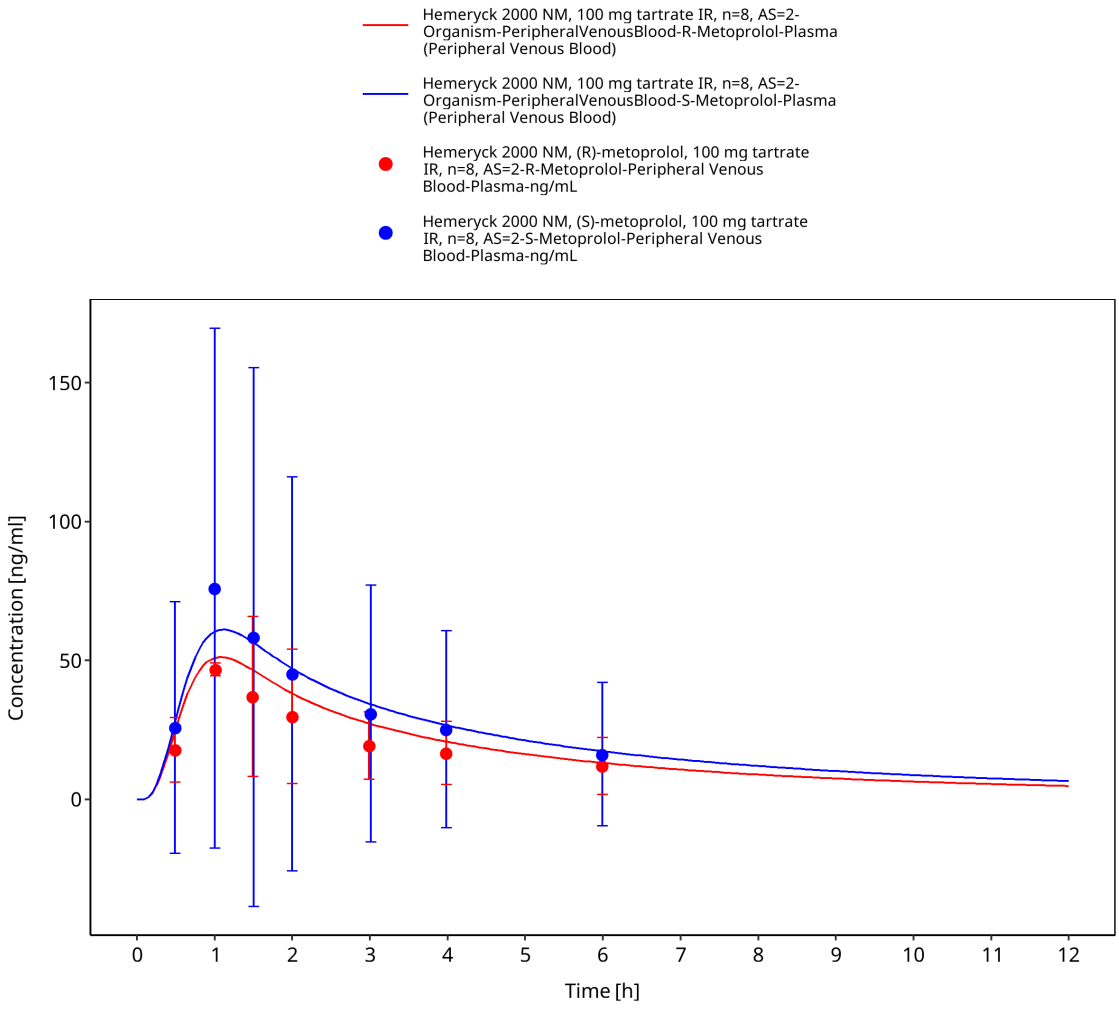


Figure 3-54: Hemeryck 2000: NM, 100 mg tartrate IR, n=8, AS=2

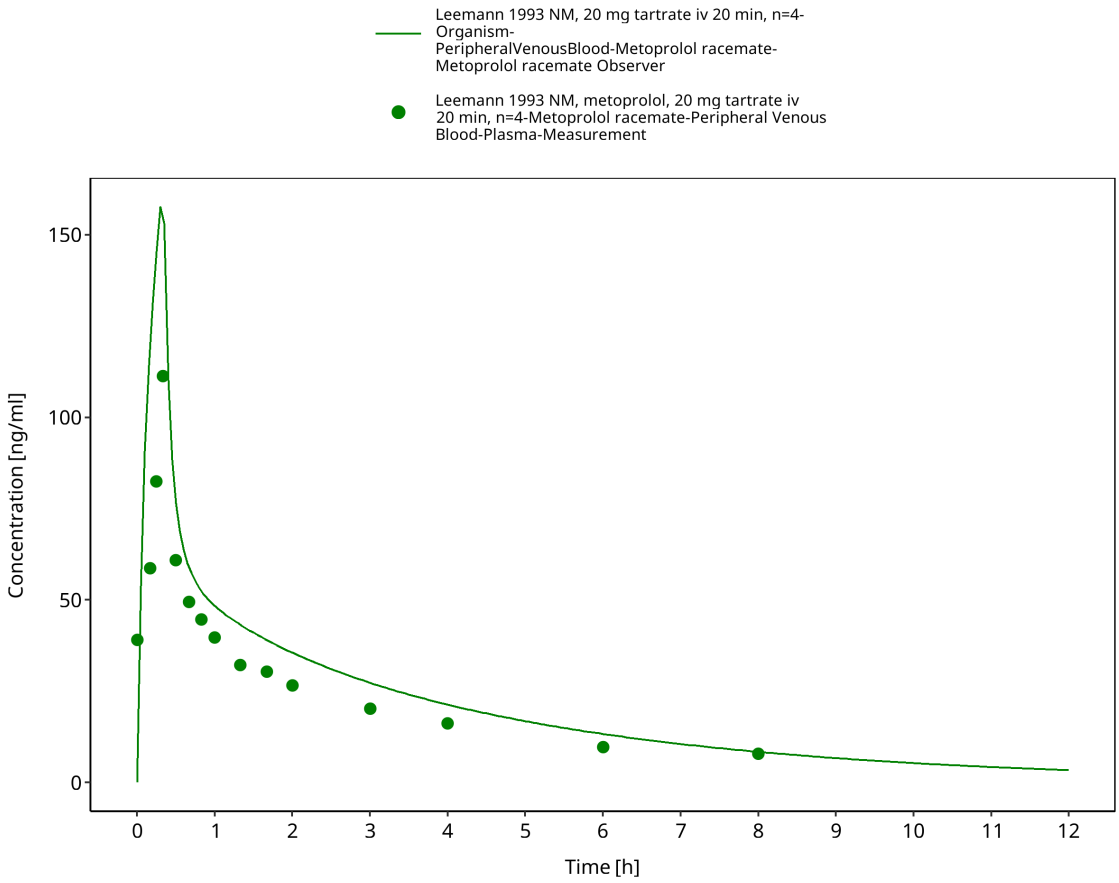


Figure 3-55: Leemann 1993: NM, 20 mg tartrate iv 20 min, n=4

4 Conclusion

The metoprolol PBPK model was evaluated against adult plasma concentration-time data for racemic metoprolol, both enantiomers and α-hydroxymetoprolol after intravenous and oral administration. The evaluated studies include immediate-release and controlled-release formulations and CYP2D6 phenotype or activity-score groups.

The report characterizes model performance only for these evaluated conditions. Material limitations include unresolved genotype differences within activity scores, fixed CYP2D6 Michaelis constants across scores and uncertainty in non-CYP2D6 metabolism.

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